Collision Ideals and Off-Diagonal Sheets

Chloe van der Vlugt

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Abstract

We study the scheme-theoretic collision relation of a polynomial self-map, namely its fiber product over the target, and describe the geometry away from the diagonal through boundary-supported inertia and Galois-closure sheets. For a polynomial map $F = (F_1, \dots, F_n)\colon \mathbb{A}^n_{\mathbb{C}} \to \mathbb{A}^n_{\mathbb{C}}$, define in $S = \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n]$ the collision and diagonal ideals

$$I_R = \big(F_i(x) - F_i(y)\big)_{i=1}^n, \qquad I_\Delta = (x_i - y_i)_{i=1}^n.$$

The image of I_Δ in the collision algebra $C_F := S/I_R$ is the obstruction ideal $\mathrm{Obs}(F) := I_\Delta/I_R$; it records the failure of the collision scheme $\operatorname{Spec} C_F$ to coincide with the diagonal. Its vanishing characterizes polynomial automorphisms:

$$\mathrm{Obs}(F) = 0 \quad \Longleftrightarrow \quad F \text{ is a polynomial automorphism.}$$

In dimension two we give two module-theoretic formulations of the same fixed-map conclusion. Planar boundary coherence asks that the boundary local cohomology $H^1_{\partial X}(\overline{X}, \mathcal{O}_{\overline{X}})$ be a finite $\overline{A}$-module; it forces $\partial X = \varnothing$, hence makes the Keller map finite étale and therefore an automorphism. Planar ramification rigidity is the divisorial criterion that $\Omega_{A_N/B}$ have no height-one support, equivalently that it have finite length. On the common Galois normalization $Z = \operatorname{Spec} A_N$, we construct the pulled-back affine opens U_g, their canonical boundary ideals $\mathfrak{j}_g = I_{A_N}(Z \setminus U_g)$, and the fixed-locus ideals $\mathfrak{a}_C$. The height-one dictionary identifies moving-sheet coverage and finite-group boundary separation as exact criteria for ramification rigidity, after which purity and finite-étale rigidity give $N = L = K$ and collision vanishing. Although $\overline{A}$ and A_N are already finite over B, neither of these additional finiteness criteria follows formally. Universal validity of either criterion is equivalent to the two-dimensional Jacobian conjecture and is not proved here. The boundary divisor-class argument does, however, prove the normal/Galois case and therefore excludes generic degree two. For the separable nonnormal cubic class in dimension three, let

$$K = \mathbb{C}(F_1, F_2, F_3), \qquad L = \mathbb{C}(x_1, x_2, x_3),$$

and let N be the normal closure of L/K. Then

$$L \otimes_K L \simeq L \times N, \qquad \mathrm{Gal}(N/K) \simeq S_3.$$

The generic off-diagonal factor N is the ordered-root Galois-closure sheet. Consequently, $\mathrm{Obs}(F) \neq 0$. The algebraic construction, the pulled-back boundary dictionary, and the implications from supplied coverage or separation hypotheses are formalized in Lean 4. The boundary-coherence and finite-length support arguments remain explicit Lean interfaces whose manuscript proofs have not yet been formalized, as do the standard external geometric statements.

Contents

1 Introduction

The Jacobian conjecture asks whether a polynomial self-map of affine space with constant nonzero Jacobian determinant must be a polynomial automorphism. Dimension two remains open. Rather than treating injectivity as an external property of the map, this paper studies the scheme-theoretic fiber product of the map with itself and asks when its off-diagonal part disappears.

There is one divisorial endgame, expressed in several exact languages. The paper does not prove the Jacobian conjecture. It organizes the remaining problem around the following two module-theoretic formulations:

$$\begin{aligned}
\mathrm{PBC}(F) &:\Longleftrightarrow H^1_{\partial X}(\overline{X}, \mathcal{O}_{\overline{X}}) \text{ is a finite } \overline{A}\text{-module},\\
\mathrm{PRR}(F) &:\Longleftrightarrow \Omega_{A_N/B} \text{ has no height-one support}\\
&\qquad\quad \Longleftrightarrow \Omega_{A_N/B} \text{ has finite length over } A_N.
\end{aligned}$$

Planar boundary coherence $\mathrm{PBC}(F)$ is the corresponding global properness formulation:

$$\mathrm{PBC}(F) \Longrightarrow \partial X = \varnothing \Longrightarrow f_F \text{ finite étale} \Longrightarrow F \text{ an automorphism.}$$

Planar ramification rigidity $\mathrm{PRR}(F)$ is the differential-module formulation of the codimension-one premise:

$$\mathrm{PRR}(F) \Longrightarrow Z/Y \text{ finite étale} \Longrightarrow N = K \Longrightarrow L = K \Longrightarrow q_F = 0 \Longrightarrow I_R = I_\Delta.$$

The divisorial endgame is controlled by an exact codimension-one obstruction on the Galois normalization. A ramified divisor E is eliminated as soon as one conjugate sheet moved by I_E still contains its generic point. In coordinates this is the conjugate-integrality condition

$$I_E \neq 1 \Longrightarrow \exists g \in G, \quad I_E \nsubseteq gHg^{-1}, \qquad w_E(g(x_1)), w_E(g(x_2)) \geq 0.$$

We call this planar moving-sheet coverage. Its ideal-theoretic form says that the fixed locus of a prime-order subgroup and the boundaries of all the sheets it moves have no common height-one component. Establishing this universally—equivalently, establishing planar ramification rigidity universally—is equivalent to $JC(2)$, so the reformulation locates the missing theorem without assuming it.

Only after all height-one points have been shown unramified do purity and finite-étale rigidity enter: they make the normal closure trivial and collapse the collision relation to the diagonal. This order is essential to the argument. The paper also develops boundary divisor-class and principal-parts criteria that may imply coverage under additional hypotheses; these are prospective mechanisms, not additional endgames or proofs of universal coverage. The divisor-class mechanism does prove unconditionally that a normal—hence Galois—marked extension L/K is trivial. In particular, a hypothetical planar counterexample must have L/K nonnormal and generic degree at least three.

The normalization rings $\overline{A}$ and A_N are finite B-modules from the outset. This makes $\overline{X}$ and Z affine varieties: they are integral affine schemes of finite type over $\mathbb{C}$. It does not make A finite over B, make boundary local cohomology finite, or make $\Omega_{A_N/B}$ finite length. Likewise, the vanishing of ordinary higher quasi-coherent cohomology on an affine scheme does not imply the vanishing or finiteness of local cohomology with support in its deleted boundary.

In the separable nonnormal cubic class in dimension three, the opposite phenomenon is visible generically: the off-diagonal factor is the ordered-root S_3-normal-closure sheet. This gives a structural comparison with the planar criteria while keeping its assumptions explicit.

Section 2 develops collision ideals and the dimension-independent normalization diagram. Section 3 constructs the planar pulled-back boundaries, proves the valuation dictionary, and states the equivalent coverage and separation criteria before applying the endgame. Section 4 treats the cubic S_3 class.

2 General construction

The organizing question is how the failure of a polynomial self-map to be one-to-one is encoded by the part of its fiber square lying off the diagonal. The collision ideal first isolates that part scheme-theoretically; the normalization diagram then describes it through conjugate sheets and inertia. These constructions are dimension-independent. Section 3 specializes them to planar Keller maps and locates the obstruction at the deleted normalization boundary, while Section 4 identifies the generic off-diagonal sheet in the separable nonnormal cubic class with an S_3-normal closure. The two specializations respectively describe when the collision scheme collapses to the diagonal and what its surviving generic sheet looks like when it does not.

2.1 The two ideals and their obstruction

Throughout the paper, the base field is $\mathbb{C}$. Let

$$F = (F_1, \dots, F_n) \colon X = \mathbb{A}^n_{\mathbb{C}} \longrightarrow \mathbb{A}^n_{\mathbb{C}}$$

be a polynomial map, and let

$$S = \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n].$$

The map F is Keller if

$$\det JF \in \mathbb{C}^{\times}.$$

Define

$$I_R(F) = \big(F_i(x) - F_i(y)\big)_{i=1}^n, \qquad I_\Delta = (x_i - y_i)_{i=1}^n, \qquad \mathrm{Obs}(F) := I_\Delta / I_R(F).$$

The quotient $\mathrm{Obs}(F)$ is naturally an ideal of the collision ring $C_F := S/I_R(F)$; we call it the obstruction ideal. For a fixed map F, we abbreviate $I_R(F)$ to I_R.

Proposition 2.1 (Canonical containment)**.** *For every polynomial map F,*

$$I_R(F) \subseteq I_\Delta.$$

Proof. For $H \in \mathbb{C}[x_1, \dots, x_n]$, the difference $H(x) - H(y)$ belongs to $(x_1 - y_1, \dots, x_n - y_n)$. This follows by induction on H, or by telescoping each monomial one coordinate at a time. Apply the statement to each coordinate F_i. □

Let $A := \mathbb{C}[x_1, \dots, x_n]$. Diagonal evaluation induces a surjective ring map

$$\bar{\mu}_F \colon C_F \longrightarrow A, \qquad [s]_{I_R} \longmapsto s(x, x).$$

Proposition 2.2 (The diagonal kernel)**.** *As ideals of C_F,*

$$\ker(\bar{\mu}_F) = \mathrm{Obs}(F) = I_\Delta / I_R.$$

Consequently,

$$\ker(\bar{\mu}_F) = 0 \iff \mathrm{Obs}(F) = 0 \iff I_R = I_\Delta.$$

Proof. The kernel of diagonal evaluation $S \to A$ is I_Δ. Passing through S/I_R therefore identifies the displayed kernel with I_Δ / I_R. Since $I_R \subseteq I_\Delta$,

$$I_\Delta / I_R = 0 \iff I_R = I_\Delta.$$

□

2.2 The fiber-product interpretation

Let

$$A = \mathbb{C}[x_1, \ldots, x_n], \qquad B = \mathbb{C}[F_1, \ldots, F_n] \subseteq A.$$

Let $X = \operatorname{Spec} A$, let $Y = \operatorname{Spec} B$, and let

$$f_F \colon X \longrightarrow Y$$

be the morphism induced by $B \hookrightarrow A$. The homomorphisms $\iota_x, \iota_y \colon A \rightrightarrows C_F$, defined by $\iota_x(a) = [a(x)]$ and $\iota_y(a) = [a(y)]$, agree on B because $[F_i(x)] = [F_i(y)]$ in C_F. Their common restriction equips C_F with a B-algebra structure.

Theorem 2.3 (Fiber-product interpretation)**.** *There is a canonical B-algebra equivalence and its dual affine-scheme equivalence*

$$C_F \cong A \otimes_B A, \qquad R_F := \operatorname{Spec} C_F \cong X \times_Y X.$$

Under the first equivalence, $\bar{\mu}_F$ is tensor multiplication

$$A \otimes_B A \longrightarrow A, \qquad a \otimes a' \longmapsto aa'.$$

Its dual morphism is the diagonal

$$\Delta_{X/Y} \colon X \longrightarrow X \times_Y X.$$

Proof. A map $C_F \to T$ to a commutative $\mathbb{C}$-algebra T is equivalent to a pair of maps $f, g \colon A \rightrightarrows T$ satisfying

$$f(F_i) = g(F_i) \qquad (1 \leq i \leq n).$$

Since the F_i generate B, this is precisely the universal property of $A \otimes_B A$. Applying Spec to this equivalence and using

$$\operatorname{Spec}(A \otimes_B A) \cong \operatorname{Spec} A \times_{\operatorname{Spec} B} \operatorname{Spec} A$$

gives the asserted fiber-product equivalence. The statements about multiplication and the diagonal follow on pure tensors. □

$$R_F(\mathbb{C}) = \{(u, v) \in X(\mathbb{C}) \times X(\mathbb{C}) : F(u) = F(v)\}.$$

Inside

$$X \times X = \operatorname{Spec} S,$$

the ideals I_R and I_Δ define, respectively, $X \times_Y X$ and the image of $\Delta_{X/Y}$. The contravariance of $I \mapsto V(I)$ therefore gives the equivalences

$$I_R \subseteq I_\Delta \iff \Delta_{X/Y}(X) \subseteq X \times_Y X$$

and

$$I_R = I_\Delta \iff X \times_Y X = \Delta_{X/Y}(X)$$

as closed subschemes of $X \times X$. Thus the algebraic and geometric gaps are two descriptions of the same scheme-theoretic containment.

2.3 The off-diagonal factor and secant projectors

For every polynomial map F, define

$$I_{\mathrm{off}} := I_R : I_\Delta, \qquad C_F^\circ := S/I_{\mathrm{off}}, \qquad R_F^\circ := \operatorname{Spec} C_F^\circ.$$

This definition does not require the diagonal to be clopen.

Corollary 2.4 (Off-diagonal emptiness criterion). *For every polynomial map F, the following are equivalent:*

$$R_F^\circ = \varnothing \iff C_F^\circ = 0 \iff \operatorname{Obs}(F) = 0 \iff I_R = I_\Delta.$$

Proof.

$$R_F^\circ = \varnothing \iff C_F^\circ = 0 \iff I_R : I_\Delta = S.$$

Moreover,

$$I_R : I_\Delta = S \iff I_\Delta \subseteq I_R \iff I_R = I_\Delta.$$

The equivalence with $\operatorname{Obs}(F) = 0$ is Proposition 2.2. □

The idempotent mechanism behind a clopen diagonal is purely commutative-algebraic. Let C be a commutative ring, $J \subseteq C$ an ideal, $\delta \in C$, and $c \in C^\times$. Suppose

$$\delta J = 0, \qquad \delta - c \in J.$$

Define

$$q := 1 - c^{-1}\delta.$$

Theorem 2.5 (Abstract off-diagonal projector). *The element q is idempotent and*

$$J = Cq, \qquad \operatorname{Ann}_C(J) = C(1-q).$$

In particular,

$$J = 0 \iff q = 0.$$

Proof. The congruence $\delta \equiv c \pmod{J}$ gives $q \in J$. For every $j \in J$,

$$qj = j - c^{-1}\delta j = j.$$

Taking $j = q$ proves $q^2 = q$; the same identity gives $J = Cq$. Finally,

$$aq = 0 \iff a = a(1-q),$$

so $\operatorname{Ann}_C(J) = \operatorname{Ann}_C(Cq) = C(1-q)$. □

Whenever the theorem is applied with

$$C = C_F, \qquad J = \operatorname{Obs}(F),$$

the identities $J = C_F q$ and $C_F/J \cong A$ give

$$C_F \cong A \times C_F/C_F(1-q).$$

Thus the first factor is the diagonal and the second is its off-diagonal complement.

2.4 Generic fibers and marked-root decomposition

Let F be dominant and generically finite, and let

$$A = \mathbb{C}[x_1, \dots, x_n], \qquad B = \mathbb{C}[F_1, \dots, F_n] \subseteq A.$$

Let

$$K = \mathrm{Frac}(B), \qquad L = \mathrm{Frac}(A).$$

Let

$$\theta \colon K \otimes_B A \longrightarrow L, \qquad \lambda \otimes a \longmapsto \lambda a.$$

Theorem 2.6 (Generic collision bridge)**.** *The map θ is an isomorphism. Consequently, there is a K-algebra isomorphism*

$$K \otimes_B C_F \cong L \otimes_K L,$$

and the following square commutes:

$$\begin{array}{ccc} K \otimes_B C_F & \xrightarrow{1 \otimes \bar{\mu}_F} & K \otimes_B A \\ \sim \downarrow & & \downarrow \theta \sim \\ L \otimes_K L & \xrightarrow{\mu_L} & L. \end{array}$$

Proof. Let $U = B \setminus \{0\}$. Since $K = U^{-1}B$, localization gives

$$K \otimes_B A \xrightarrow{\sim} U^{-1}A, \qquad \frac{b}{u} \otimes a \longmapsto \frac{ba}{u}.$$

Since $A = \mathbb{C}[x_1, \dots, x_n]$ is a domain and $B \subseteq A$,

$$B \setminus \{0\} \subseteq A \setminus \{0\}.$$

Hence

$$U^{-1}A \hookrightarrow \mathrm{Frac}(A) = L, \qquad \frac{a}{u} \longmapsto \frac{a}{u}.$$

Composing this injection with the displayed isomorphism gives θ, so θ is injective.

Since F is generically finite, L/K is a finite extension. The image of $U^{-1}A$ in L is

$$K[x_1, \dots, x_n].$$

Since each x_i is algebraic over K, this K-domain is finite-dimensional over K, hence is a field. Since it contains A, it contains $\mathrm{Frac}(A) = L$; the reverse inclusion is already given by its embedding into L. Therefore

$$U^{-1}A = L,$$

and θ is an isomorphism.

By Theorem 2.3, identify C_F with $A \otimes_B A$. There is a K-algebra isomorphism

$$\beta \colon K \otimes_B (A \otimes_B A) \xrightarrow{\sim} (K \otimes_B A) \otimes_K (K \otimes_B A)$$

defined on pure tensors by

$$\beta(\lambda \otimes (a \otimes a')) = (\lambda \otimes a) \otimes (1 \otimes a').$$

Its inverse is

$$(\lambda \otimes a) \otimes (\lambda' \otimes a') \longmapsto \lambda\lambda' \otimes (a \otimes a').$$

The balancing relations over B and K show that these formulas are well-defined, and the two displayed maps are mutually inverse. Composing β with $\theta \otimes_K \theta$ gives

$$\Phi_F \colon K \otimes_B C_F \xrightarrow{\sim} L \otimes_K L.$$

It remains to identify the diagonal map. Under $C_F \cong A \otimes_B A$, the homomorphism $\bar{\mu}_F$ is multiplication. Therefore, on a pure tensor,

$$(\mu_L \circ \Phi_F)\big(\lambda \otimes (a \otimes a')\big) = (\lambda a)a' = \lambda a a',$$
$$\big(\theta \circ (1 \otimes \bar{\mu}_F)\big)\big(\lambda \otimes (a \otimes a')\big) = \theta(\lambda \otimes aa') = \lambda a a'.$$

Pure tensors generate $K \otimes_B C_F$, so the two homomorphisms are equal. This proves the commutative square. □

Suppose now that L/K is separable and has a power basis $L = K(\alpha)$. Let $m_\alpha(T) \in K[T]$ be the minimal polynomial of α, and let $h_\alpha(T) \in L[T]$ be determined by

$$m_\alpha(T) = (T - \alpha)h_\alpha(T) \qquad \text{in } L[T].$$

Theorem 2.7 (Marked-root tensor decomposition)**.** *Give $L \otimes_K L$ its L-algebra structure through the left tensor factor. There is an L-algebra equivalence*

$$L \otimes_K L \cong L \times L[T]/(h_\alpha),$$

under which tensor multiplication $L \otimes_K L \to L$ is projection onto the first factor.

Proof. The power-basis presentation

$$L \cong K[T]/(m_\alpha)$$

and base change from K to L give an L-algebra isomorphism

$$L \otimes_K L \xrightarrow{\sim} L[T]/(m_\alpha), \qquad 1 \otimes \alpha \longmapsto T.$$

Since m_α is separable, $m_\alpha'(\alpha) \neq 0$. Differentiating

$$m_\alpha(T) = (T - \alpha)h_\alpha(T)$$

and evaluating at $T = \alpha$ gives

$$h_\alpha(\alpha) = m_\alpha'(\alpha) \neq 0.$$

Thus $T - \alpha$ does not divide h_α, so these two polynomials are coprime in $L[T]$. The Chinese remainder theorem now gives

$$L[T]/(m_\alpha) \xrightarrow{\sim} L[T]/(T - \alpha) \times L[T]/(h_\alpha) \xrightarrow{\sim} L \times L[T]/(h_\alpha).$$

Under the first isomorphism above, a pure tensor $\ell \otimes p(\alpha)$, with $p \in K[T]$, corresponds to the class of $\ell p(T)$. Tensor multiplication sends it to $\ell p(\alpha)$, which is evaluation at $T = \alpha$. Under the Chinese remainder isomorphism, this evaluation map is projection onto the factor $L[T]/(T - \alpha) \cong L$. Hence tensor multiplication is the first projection. □

Together with Theorem 2.6, this gives

$$K \otimes_B C_F \cong L \times L[T]/(h_\alpha),$$

where the first projection is the generic diagonal.

2.5 Normalized covers and inertia

Let L/K be finite and separable, and let N/K be its normal closure, with a fixed embedding $L \hookrightarrow N$. Let

$$G = \mathrm{Gal}(N/K), \qquad H = \mathrm{Gal}(N/L).$$

Proposition 2.8 (Conjugate-embedding action)**.** *Restriction induces a bijection*

$$G/H \xrightarrow{\sim} \mathrm{Hom}_K(L, N), \qquad gH \longmapsto g|_L.$$

The induced action of G on G/H is faithful; equivalently,

$$\mathrm{core}_G(H) = \bigcap_{g \in G} gHg^{-1} = 1.$$

Proof. For $g_1, g_2 \in G$,

$$g_1|_L = g_2|_L \quad \Longleftrightarrow \quad g_2^{-1} g_1 \in H \quad \Longleftrightarrow \quad g_1 H = g_2 H.$$

Every K-embedding $L \hookrightarrow N$ extends to an element of G, so restriction gives the stated bijection. Its action kernel is

$$\ker(G \curvearrowright G/H) = \bigcap_{g \in G} gHg^{-1} = \mathrm{core}_G(H).$$

If $\sigma \in \mathrm{core}_G(H)$, then

$$\sigma|_{g(L)} = \mathrm{id}_{g(L)} \quad (g \in G).$$

Since N is the normal closure,

$$N = K\big(g(L) : g \in G\big),$$

and therefore $\sigma = \mathrm{id}_N$. □

Let

$$Y = \operatorname{Spec} B, \qquad \overline{X} = \mathrm{Norm}_L(Y), \qquad Z = \mathrm{Norm}_N(Y).$$

The marked embedding $L \hookrightarrow N$ induces a canonical commutative triangle

$$Z \xrightarrow{\pi} \overline{X} \xrightarrow{\bar{\nu}} Y, \qquad \nu = \bar{\nu} \circ \pi \colon Z \longrightarrow Y.$$

Suppose that $\bar{\nu}$ and ν are finite, and fix an open immersion over Y

$$j \colon X = \operatorname{Spec} A \hookrightarrow \overline{X}$$

with $\bar{\nu} \circ j = f_F$. Denote this diagram and its deleted boundary by

$$\mathcal{D} = \Big(Z \xrightarrow{\pi} \overline{X} \xrightarrow{\bar{\nu}} Y,\, j \colon X \hookrightarrow \overline{X}\Big), \qquad \partial X = \overline{X} \setminus j(X).$$

Fix $E \in Z^{(1)}$. Let w_E be its divisorial valuation, let $b = \nu(E)$, and let $v_b = w_E|_K$. Let $D_E \leq G$ and $I_E \trianglelefteq D_E$ be its decomposition and inertia groups. Let

$$\mathcal{Q}_E := D_E \backslash G/H.$$

For $\omega = D_E g H \in \mathcal{Q}_E$, let

$$u_\omega = (g^{-1} w_E)|_L, \qquad \mathrm{cent}(E, \omega) = \mathrm{cent}_{\overline{X}}(u_\omega),$$

and let

$$i_E(\omega) = [I_E : I_E \cap gHg^{-1}].$$

Proposition 2.9 (Conjugate divisors and double cosets)**.** *The assignments above are well-defined, and restriction of valuations induces a bijection*

$$\mathcal{Q}_E \xrightarrow{\sim} \{\textit{prime divisors of } \overline{X} \textit{ above } b\}, \qquad \omega \longmapsto \mathrm{cent}(E,\omega).$$

Proof. The group G acts transitively on the extensions of v_b to N, and the stabilizer of w_E is D_E. Restricting such a valuation to $L = N^H$ identifies precisely those extensions whose indices differ on the right by an element of H. The resulting orbits are therefore indexed by $D_E\backslash G/H$, giving the stated bijection.

It remains to check independence of representatives. If $D_E g' H = D_E g H$, then $g' = dgh$ for some $d \in D_E$ and $h \in H$. The element d fixes w_E, while h fixes L pointwise, so

$$(g'^{-1}w_E)|_L = (g^{-1}w_E)|_L.$$

Moreover, $I_E \trianglelefteq D_E$, and conjugation by d identifies

$$I_E \cap g'Hg'^{-1} \quad \text{with} \quad I_E \cap gHg^{-1}.$$

Thus u_ω, $\mathrm{cent}(E,\omega)$, and $i_E(\omega)$ are well-defined. □

We use the notation

$$e\big(\mathrm{cent}(E,\omega)/b\big) := e(u_\omega/v_b).$$

Since the residue characteristic is zero,

$$\mathrm{Ram}^{(1)}(Z/Y) := \{E \in Z^{(1)} : e(w_E/v_b) > 1\} = \{E \in Z^{(1)} : I_E \neq 1\}.$$

Proposition 2.10 (Detection of inertia on a conjugate sheet)**.** *If $I_E \neq 1$, then there is an $\omega \in \mathcal{Q}_E$ with*

$$i_E(\omega) > 1.$$

Proof. Suppose, to the contrary, that $i_E(\omega) = 1$ for every $\omega \in \mathcal{Q}_E$. For each $g \in G$,

$$1 = i_E(D_E gH) = [I_E : I_E \cap gHg^{-1}],$$

so

$$I_E = I_E \cap gHg^{-1} \subseteq gHg^{-1} \qquad \text{for every } g \in G.$$

Consequently,

$$I_E \subseteq \bigcap_{g\in G} gHg^{-1} = \mathrm{core}_G(H) = 1,$$

contrary to $I_E \neq 1$. □

Proposition 2.11 (Intermediate ramification index)**.** *For $\omega = D_E gH \in \mathcal{Q}_E$,*

$$e\big(\mathrm{cent}(E,\omega)/b\big) = i_E(\omega) = [I_E : I_E \cap gHg^{-1}].$$

Proof. Let w_E and v_b be the divisorial valuations associated with E and b, and let

$$w_g = g^{-1}w_E, \qquad u_\omega = w_g|_L.$$

The valuation u_ω has center $\mathrm{cent}(E,\omega)$ on $\overline{X}$. Since the residue fields have characteristic zero,

$$e(w_g/v_b) = |I_E|.$$

The inertia group of w_g over u_ω is

$$g^{-1}I_E g \cap H,$$

and hence

$$e(w_g/u_\omega) = |g^{-1}I_E g \cap H| = |I_E \cap gHg^{-1}|.$$

Multiplicativity of ramification indices in the field tower

$$K \subseteq L \subseteq N$$

therefore gives

$$e(u_\omega/v_b) = \frac{e(w_g/v_b)}{e(w_g/u_\omega)} = [I_E : I_E \cap gHg^{-1}],$$

which is the asserted formula. $\square$

Proposition 2.12 (Visibility-to-boundary)**.** *Suppose that $X \to Y$ is étale. For $E \in Z^{(1)}$ and $\omega \in \mathcal{Q}_E$,*

$$i_E(\omega) > 1 \quad \Longrightarrow \quad \mathrm{cent}(E,\omega) \in \partial X.$$

Proof. Suppose that $\mathrm{cent}(E,\omega) \in j(X)$, and let $x \in X$ be the corresponding point under the open immersion j. The local homomorphism

$$\mathcal{O}_{Y,b} \longrightarrow \mathcal{O}_{X,x}$$

is étale, so $e(\mathrm{cent}(E,\omega)/b) = 1$. Proposition 2.11 now gives

$$i_E(\omega) = e\big(\mathrm{cent}(E,\omega)/b\big) = 1.$$

Taking the contrapositive gives the result. $\square$

Definition 2.13 (Hidden-inertia locus)**.**

$$\mathcal{H}(\mathcal{D}) := \left\{ E \in \mathrm{Ram}^{(1)}(Z/Y) : \begin{array}{l} \exists \omega \in \mathcal{Q}_E,\ i_E(\omega) > 1, \\ \forall \omega \in \mathcal{Q}_E, \quad i_E(\omega) > 1 \Longrightarrow \mathrm{cent}(E,\omega) \in \partial X \end{array} \right\}.$$

Thus $E \in \mathcal{H}(\mathcal{D})$ precisely when nontrivial inertia is visible on a conjugate sheet but every visible center belongs to the deleted boundary.

Proposition 2.14 (Divisorial ramification is hidden from the étale sheet)**.** *Suppose that N/K is the normal closure of L/K, the residue characteristic is zero, and $X \to Y$ is étale. Then*

$$\mathcal{H}(\mathcal{D}) = \mathrm{Ram}^{(1)}(Z/Y).$$

Equivalently, every divisorial ramification orbit has a conjugate sheet on which its inertia acts nontrivially, but every such center lies in the deleted boundary.

Proof. The inclusion $\mathcal{H}(\mathcal{D}) \subseteq \mathrm{Ram}^{(1)}(Z/Y)$ is part of the definition. Conversely, let $E \in \mathrm{Ram}^{(1)}(Z/Y)$. In residue characteristic zero, $I_E \neq 1$. Since N is the normal closure of L/K, Proposition 2.8 gives $\mathrm{core}_G(H) = 1$, and Proposition 2.10 supplies an $\omega \in \mathcal{Q}_E$ with $i_E(\omega) > 1$. For every $\omega' \in \mathcal{Q}_E$, Proposition 2.12 gives

$$i_E(\omega') > 1 \quad \Longrightarrow \quad \mathrm{cent}(E,\omega') \in \partial X.$$

These are exactly the two conditions defining $E \in \mathcal{H}(\mathcal{D})$. $\square$

2.6 Automorphism detection

Theorem 2.15 (Automorphism criterion)**.** *For an arbitrary polynomial self-map* $F\colon \mathbb{A}^n_{\mathbb{C}} \to \mathbb{A}^n_{\mathbb{C}}$*, the following are equivalent:*

(1) *F is a polynomial automorphism;*

(2) *F is injective on $\mathbb{A}^n(\mathbb{C})$;*

(3) $I_R(F) = I_\Delta$*;*

(4) $\mathrm{Obs}(F) = 0$*.*

Proof. Suppose first that F has polynomial inverse $G = (G_1, \dots, G_n)$. Then

$$x_i - y_i = G_i(F(x)) - G_i(F(y)) \in I_R$$

for every i, so $I_\Delta \subseteq I_R$. Proposition 2.1 gives equality. Thus (1) implies (3).

Suppose $I_R = I_\Delta$, and let $\mathrm{ev}_{(a,b)}\colon S \to \mathbb{C}$ be evaluation at (a, b). If $F(a) = F(b)$, then

$$I_R \subseteq \ker(\mathrm{ev}_{(a,b)}).$$

Therefore $x_i - y_i \in I_\Delta = I_R$ gives $a_i - b_i = 0$ for every i. Thus $a = b$, proving (3)⇒(2). The Ax–Grothendieck theorem [Ax69] gives (2)⇒(1). Finally, Proposition 2.2 gives (3)⟺(4). ☐

Corollary 2.16 (The Jacobian conjecture as obstruction vanishing)**.** *For every $n \geq 1$, the n-dimensional Jacobian conjecture is equivalent to each of the statements*

$$\forall F\colon \mathbb{A}^n_{\mathbb{C}} \to \mathbb{A}^n_{\mathbb{C}}, \qquad \det JF \in \mathbb{C}^\times \quad \Longrightarrow \quad \mathrm{Obs}(F) = 0,$$

and

$$\forall F\colon \mathbb{A}^n_{\mathbb{C}} \to \mathbb{A}^n_{\mathbb{C}}, \qquad \det JF \in \mathbb{C}^\times \quad \Longrightarrow \quad I_R(F) = I_\Delta.$$

Proof. Apply Theorem 2.15 to every Keller map. Its equivalent conditions (1), (3), and (4) give the two displayed formulations. ☐

2.7 Generic degree one

Corollary 2.17 (Generic degree-one descent)**.** *If*

$$L = K \qquad \textit{and} \qquad A \textit{ is flat over } B,$$

then

$$\mathrm{Obs}(F) = 0.$$

Proof. When $L = K$, Theorem 2.6 identifies the base-changed diagonal with tensor multiplication:

$$\ker(1 \otimes \bar{\mu}_F) \cong \ker\Big(K \otimes_K K \xrightarrow{\mu_K} K\Big) = 0.$$

The fiber-product identification of Theorem 2.3 is

$$C_F \xrightarrow{\sim} A \otimes_B A.$$

Since A is flat over B, the tensor product $A \otimes_B A$, and therefore C_F, is flat over B. The localization map

$$\iota_C \colon C_F \hookrightarrow K \otimes_B C_F$$

is therefore injective, and

$$\iota_C(\ker \bar{\mu}_F) \subseteq \ker(1 \otimes \bar{\mu}_F) = 0.$$

Thus $\ker \bar{\mu}_F = 0$. Proposition 2.2 identifies

$$\ker(\bar{\mu}_F) = \mathrm{Obs}(F) = I_\Delta / I_R,$$

so $\mathrm{Obs}(F) = 0$. □

Lean correspondence. `CollisionIdeals/CollisionDiagonal.lean`, `CollisionIdeals/OffDiagonalScheme.lean`, `CollisionIdeals/Secant.lean`, `CollisionIdeals/GenericCollisionDiagonal.lean`, `CollisionIdeals/KellerCollisionModel.lean`, and `CollisionIdeals/JacobianConjecture.lean`. The Lean form of Theorem 2.6 takes surjectivity of θ as a parameter; the proof above derives it from generic finiteness.

3 Planar vanishing

We now specialize the dimension-independent collision, normalization, and inertia constructions of Section 2 to dimension two. The purpose of this section is to identify a concrete boundary obstruction, not to assume it away implicitly: the secant calculation separates the diagonal from the off-diagonal collision sheet, while the normalization calculation shows exactly how nontrivial inertia would have to escape every relevant conjugate affine sheet.

Let

$$F = (P, Q) \colon \mathbb{A}^2_{\mathbb{C}} \longrightarrow \mathbb{A}^2_{\mathbb{C}}$$

be a polynomial map satisfying $\det JF = c \in \mathbb{C}^\times$.

Set, from this point onward,

$$A = \mathbb{C}[x_1, x_2], \qquad B = \mathbb{C}[P, Q] \subseteq A, \qquad K = \mathbb{C}(P, Q), \qquad L = \mathbb{C}(x_1, x_2),$$

and let $X = \operatorname{Spec} A$, $Y = \operatorname{Spec} B$.

Proposition 3.1 (Planar Keller geometry)**.**

$$\mathbb{C}[u, v] \xrightarrow{\sim} B, \qquad u \longmapsto P, \quad v \longmapsto Q.$$

Moreover,

$$X \xrightarrow{f_F} Y \text{ is étale}, \qquad A \text{ is flat over } B, \qquad [L : K] < \infty \text{ and } L/K \text{ is separable.}$$

Proof. The identity

$$dP \wedge dQ = \left(\frac{\partial P}{\partial x_1} \frac{\partial Q}{\partial x_2} - \frac{\partial P}{\partial x_2} \frac{\partial Q}{\partial x_1} \right) dx_1 \wedge dx_2 = c\, dx_1 \wedge dx_2 \neq 0$$

shows that P, Q are algebraically independent. Thus $B \cong \mathbb{C}[u, v]$ and $Y \cong \mathbb{A}^2_{\mathbb{C}}$. Moreover,

$$A \cong \mathbb{C}[u, v, U, V] / (P(U, V) - u,\ Q(U, V) - v)$$

as a $\mathbb{C}[u, v]$-algebra, and its relative Jacobian determinant is $c \in A^\times$. The Jacobian criterion therefore makes $B \to A$ étale. In particular it is flat and quasi-finite. Since f_F is dominant and quasi-finite, its generic fiber is zero-dimensional, so $[L : K] < \infty$. The morphism remains étale at the generic point; hence the finite extension L/K is separable. □

Proposition 3.2 (Finite planar Keller rigidity)**.** *If the planar Keller morphism $f_F\colon \mathbb{A}^2_{\mathbb{C}} \to \mathbb{A}^2_{\mathbb{C}}$ is finite, then F is a polynomial automorphism.*

Proof. Proposition 3.1 makes f_F étale. Thus a finite f_F is a connected finite étale cover of $\mathbb{A}^2_{\mathbb{C}}$. Finite-étale rigidity of the affine plane, Theorem 3.28, makes it an isomorphism. Its inverse is a morphism of affine schemes and hence is given by polynomials. $\square$

Definition 3.3 (Planar generic degree)**.** The generic degree of F is

$$d_F := \deg_{\mathrm{gen}}(F) := [L : K].$$

Because L/K is finite and separable, d_F is also the number of K-embeddings $L \hookrightarrow \overline{K}$, equivalently the number of geometric points in the generic fiber of f_F.

Proposition 3.4 (Keller inverse-Jacobian frame)**.** *Define polynomial derivations of $A = \mathbb{C}[x_1, x_2]$ by*

$$\partial_P := c^{-1}\left(Q_{x_2}\frac{\partial}{\partial x_1} - Q_{x_1}\frac{\partial}{\partial x_2}\right), \qquad \partial_Q := c^{-1}\left(-P_{x_2}\frac{\partial}{\partial x_1} + P_{x_1}\frac{\partial}{\partial x_2}\right).$$

Then

$$\partial_P(P) = 1, \quad \partial_P(Q) = 0, \quad \partial_Q(P) = 0, \quad \partial_Q(Q) = 1.$$

They extend uniquely to derivations of $L = \mathrm{Frac}(A)$, and their restrictions to $K = \mathbb{C}(P, Q)$ are the coordinate derivations with respect to P, Q.

Proof. The first four identities are the inverse-matrix identity for JF, using $\det JF = c$. A derivation extends uniquely to a localization, hence to $L = \mathrm{Frac}(A)$. The four identities then give the asserted restrictions to K. $\square$

3.1 The planar secant projector

Let $\Delta = V(I_\Delta) \subseteq \operatorname{Spec} S$. For $a \in S$, write $a|_\Delta$ for its image under $S \to S/I_\Delta$, and apply this notation entrywise to matrices. Let

$$\mathbf{d} = \begin{pmatrix} x_1 - y_1 \\ x_2 - y_2 \end{pmatrix}, \qquad \mathbf{r} = \begin{pmatrix} P(x) - P(y) \\ Q(x) - Q(y) \end{pmatrix}.$$

For $H \in \mathbb{C}[x_1, x_2]$, let

$$D_1 H = \frac{H(x_1, x_2) - H(y_1, x_2)}{x_1 - y_1}, \qquad D_2 H = \frac{H(y_1, x_2) - H(y_1, y_2)}{x_2 - y_2}.$$

Both quotients belong to S. Let

$$M_F = \begin{pmatrix} D_1 P & D_2 P \\ D_1 Q & D_2 Q \end{pmatrix}.$$

Then

$$\mathbf{r} = M_F \mathbf{d}, \qquad M_F\big|_\Delta = JF.$$

Let $\delta_F = \det(M_F)$.

Proposition 3.5 (Planar secant annihilation)**.** *The secant determinant satisfies*

$$\delta_F I_\Delta \subseteq I_R, \qquad \delta_F\big|_\Delta = c.$$

Proof. Multiplying $\mathbf{r} = M_F\mathbf{d}$ by $\operatorname{adj}(M_F)$ gives

$$\delta_F\mathbf{d} = \operatorname{adj}(M_F)\mathbf{r}.$$

$$\mathbf{r} \in I_R^{\oplus 2}, \qquad \operatorname{adj}(M_F) \in M_2(S) \quad\Longrightarrow\quad \operatorname{adj}(M_F)\mathbf{r} \in I_R^{\oplus 2}.$$

Therefore

$$\delta_F\mathbf{d} \in I_R^{\oplus 2},$$

or equivalently,

$$\forall j \in \{1,2\}, \qquad \delta_F(x_j - y_j) \in I_R.$$

Hence $\delta_F I_\Delta \subseteq I_R$. On the diagonal,

$$(D_jF_i)|_\Delta = \frac{\partial F_i}{\partial x_j},$$

so $M_F|_\Delta = JF$ and $\delta_F|_\Delta = \det JF = c$. □

Let $\bar\delta_F$ be the class of δ_F in C_F, and define

$$q_F := 1 - c^{-1}\bar\delta_F.$$

Set also

$$e_F := 1 - q_F = c^{-1}\bar\delta_F.$$

Proposition 3.6 (Secant separability idempotent)**.** *Under $C_F \cong A \otimes_B A$, the element e_F satisfies*

$$e_F^2 = e_F, \qquad \bar\mu_F(e_F) = 1, \qquad (a\otimes 1)e_F = (1\otimes a)e_F \quad (a \in A).$$

Thus e_F is the diagonal separability idempotent written explicitly by the planar secant determinant.

Proof. Proposition 3.5 gives $\bar\delta_F J = 0$, while $\bar\delta_F - c \in J$. Hence $\bar\delta_F(\bar\delta_F - c) = 0$, so $e_F^2 = e_F$. Diagonal evaluation gives $\bar\mu_F(\bar\delta_F) = c$, so $\bar\mu_F(e_F) = 1$. Finally, $a\otimes 1 - 1\otimes a$ belongs to $J = \ker(\bar\mu_F)$, while e_F annihilates J; this is exactly the last displayed identity. □

Theorem 3.7 (Planar collision-sheet decomposition)**.** *The collision scheme decomposes as*

$$R_F \cong \Delta_{X/Y}(X) \amalg R_F^\circ, \qquad C_F \cong A \times C_F^\circ.$$

The off-diagonal ideal satisfies

$$I_{\text{off}} = I_R + (\delta_F) = I_R : I_\Delta^\infty.$$

Proof. By Proposition 2.2,

$$J := \ker(\bar\mu_F) = I_\Delta/I_R = \operatorname{Obs}(F).$$

Proposition 3.5 gives

$$\bar\delta_F J = 0, \qquad \bar\delta_F - c \in J.$$

Theorem 2.5, applied to these two relations and $q_F = 1 - c^{-1}\bar{\delta}_F$, yields

$$q_F^2 = q_F, \qquad J = C_F q_F, \qquad \mathrm{Ann}_{C_F}(J) = C_F(1-q_F), \qquad J = 0 \Longleftrightarrow q_F = 0.$$

Because $1 - q_F = c^{-1}\bar{\delta}_F$,

$$\frac{I_R : I_\Delta}{I_R} = \mathrm{Ann}_{C_F}(J) = C_F(1-q_F) = C_F\bar{\delta}_F = \frac{I_R + (\delta_F)}{I_R}.$$

Hence

$$I_{\mathrm{off}} = I_R : I_\Delta = I_R + (\delta_F).$$

Idempotence gives $J^m = C_F q_F = J$ for every $m \geq 1$. Therefore

$$\frac{I_R : I_\Delta^m}{I_R} = \mathrm{Ann}_{C_F}(J^m) = \mathrm{Ann}_{C_F}(J) = \frac{I_R : I_\Delta}{I_R}.$$

Thus

$$I_R : I_\Delta^\infty = \bigcup_{m\geq 1} (I_R : I_\Delta^m) = I_R : I_\Delta.$$

The idempotent q_F also gives a ring isomorphism

$$C_F \xrightarrow{\sim} C_F(1-q_F) \times C_F q_F, \qquad a \longmapsto \big(a(1-q_F), aq_F\big),$$

whose inverse is addition. The two factors satisfy

$$C_F(1-q_F) \cong C_F/C_F q_F = C_F/J \xrightarrow[\bar{\mu}_F]{\sim} A$$

and

$$C_F q_F \cong C_F/C_F(1-q_F) \cong S/\big(I_R + (\delta_F)\big) = C_F^\circ.$$

Thus $C_F \cong A \times C_F^\circ$. Since the spectrum of a product is the disjoint union of the two spectra, this isomorphism dualizes to

$$R_F \cong \Delta_{X/Y}(X) \amalg R_F^\circ.$$

$\square$

Corollary 2.4 states that

$$R_F^\circ = \varnothing \quad\Longleftrightarrow\quad C_F^\circ = 0 \quad\Longleftrightarrow\quad \mathrm{Obs}(F) = 0 \quad\Longleftrightarrow\quad I_R = I_\Delta.$$

The preceding theorem identifies

$$C_F^\circ = S/\big(I_R + (\delta_F)\big),$$

and Theorem 2.5 gives $\mathrm{Obs}(F) = 0 \Longleftrightarrow q_F = 0$. Hence

$$q_F = 0 \quad\Longleftrightarrow\quad I_R + (\delta_F) = S \quad\Longleftrightarrow\quad R_F^\circ = \varnothing \quad\Longleftrightarrow\quad I_R = I_\Delta.$$

In particular, the secant determinant cuts out the off-diagonal factor:

$$R_F^\circ = \mathrm{Spec}\big(S/(I_R + (\delta_F))\big).$$

The entries of the full secant matrix give more than its determinant. Write

$$M_F = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix}, \qquad d_i = x_i - y_i, \qquad r_1 = P(x) - P(y), \quad r_2 = Q(x) - Q(y),$$

and define

$$\Phi_F = \begin{pmatrix} m_{22} & -m_{21} \\ -m_{12} & m_{11} \\ -d_1 & -d_2 \end{pmatrix}.$$

Proposition 3.8 (Secant Hilbert–Burch resolution)**.** *Assume that $C_F^\circ \neq 0$. The signed maximal minors of Φ_F are r_1, r_2, δ_F, and there is an exact sequence*

$$0 \longrightarrow S^2 \xrightarrow{\Phi_F} S^3 \xrightarrow{(r_1, r_2, \delta_F)} S \longrightarrow C_F^\circ \longrightarrow 0.$$

Consequently

$$\omega_{C_F^\circ} \cong \operatorname{coker}(\Phi_F^{\mathsf{t}}) \cong C_F^\circ.$$

The last isomorphism is induced by

$$(u, v) \longmapsto -d_2 u + d_1 v.$$

Proof. Direct calculation gives

$$\det\begin{pmatrix} -m_{12} & m_{11} \\ -d_1 & -d_2 \end{pmatrix} = r_1, \quad -\det\begin{pmatrix} m_{22} & -m_{21} \\ -d_1 & -d_2 \end{pmatrix} = r_2, \quad \det\begin{pmatrix} m_{22} & -m_{21} \\ -m_{12} & m_{11} \end{pmatrix} = \delta_F.$$

Thus the maximal-minor ideal is $I_R + (\delta_F) = I_{\mathrm{off}}$. The first projection makes $C_F = A \otimes_B A$ an étale A-algebra, so its nonzero direct factor C_F° is a regular Cohen–Macaulay surface. Hence $I_{\mathrm{off}} \subset S$ has grade two, and Hilbert–Burch gives the displayed resolution [The26s].

Dualizing the resolution over the regular four-dimensional ring S gives $\omega_{C_F^\circ} \cong \operatorname{coker}(\Phi_F^{\mathsf{t}})$. The row $(-d_2, d_1)$ kills every row of Φ_F modulo (r_1, r_2, δ_F), so it induces the stated map to C_F°. Moreover $(d_1, d_2)C_F^\circ = C_F^\circ$: otherwise a prime containing both would also contain δ_F, although $\delta_F \equiv c \pmod{(d_1, d_2)}$. The induced map is therefore surjective. Both source and target are rank-one locally free modules on the regular scheme $\operatorname{Spec} C_F^\circ$, so it is an isomorphism. □

The final trivialization is intrinsic up to a unit. Identifying this particular Hilbert–Burch generator with a chosen volume form requires a separate comparison; no such identification is used below.

3.2 Galois specialization and conjugate coordinates

Fix an algebraic closure $\overline{K} \supseteq L$, and let

$$\begin{aligned} N &:= K\big(\sigma(L) : \sigma \in \operatorname{Hom}_K(L, \overline{K})\big), & K \subseteq L \subseteq N \subseteq \overline{K}, \\ \overline{A} &:= \{a \in L : a \text{ is integral over } B\}, & A_N := \{a \in N : a \text{ is integral over } B\}, \\ \overline{X} &:= \operatorname{Spec} \overline{A}, & Z := \operatorname{Spec} A_N. \end{aligned}$$

Thus N/K is the normal closure of L/K.

Proposition 3.9 (Planar finite-cover diagram)**.**

$$Z \xrightarrow[\text{finite}]{\pi} \overline{X} \xrightarrow[\text{finite}]{\bar{\nu}} Y, \qquad \nu = \bar{\nu} \circ \pi,$$

and

$$X \xhookrightarrow[\text{open}]{j} \overline{X}, \qquad \bar{\nu} \circ j = f_F.$$

Proof. Since $B \cong \mathbb{C}[u, v]$ is a finitely generated normal $\mathbb{C}$-algebra, its integral closures in the finite extensions L/K and N/K are finite over B. The inclusion $L \subseteq N$ gives $\overline{A} \subseteq A_N$, so

$$B \subseteq \overline{A} \subseteq A_N.$$

The morphism $\bar{\nu}$ is finite because $\overline{A}$ is a finite B-module. If $a_1, \dots, a_r$ generate A_N over B, then they also generate A_N over $\overline{A}$, since $B \subseteq \overline{A}$. Hence π is finite.

For $a \in \overline{A}$, integrality over $B \subseteq A$ and the normality of $A = \mathbb{C}[x_1, x_2]$ give $a \in A$. Thus

$$B \subseteq \overline{A} \subseteq A,$$

and $\mathrm{Frac}(\overline{A}) = L$: clearing denominators in an algebraic equation over $K = \mathrm{Frac}(B)$ shows that for every $\ell \in L$, some nonzero $b \in B$ makes $b\ell$ integral over B. Thus $\ell = (b\ell)/b \in \mathrm{Frac}(\overline{A})$. The inclusion $\overline{A} \subseteq A$ therefore induces a dominant birational morphism $j \colon X \to \overline{X}$.

The morphism j is quasi-finite: because $\bar{\nu} \circ j = f_F$, each fiber of j is contained in a fiber of the quasi-finite morphism f_F. It is separated because it is a morphism between affine schemes. Since $\overline{X}$ is normal, Zariski's Main Theorem makes j an open immersion. □

One consequence of this finiteness is useful bookkeeping, but should not be confused with a vanishing theorem. If $E = V(\mathfrak{p}_E) \subset Z$ is a prime divisor and $D = V(\mathfrak{p}_E \cap B) \subset Y$ is its image, then

$$A_N/\mathfrak{p}_E \quad \text{is finite over} \quad B/(\mathfrak{p}_E \cap B).$$

Thus $E \to D$ is a finite map of affine curves. Neither coordinate ring is thereby finite-dimensional over $\mathbb{C}$, and this curvewise finiteness does not bound poles in the direction transverse to E. More precisely, if $e(E/D)$ is the DVR ramification index and $f(E/D) = [\kappa(E) : \kappa(D)]$ is the degree of the induced curve map, then

$$\sum_{E|D} e(E/D) f(E/D) = [N : K].$$

Riemann–Hurwitz or equal-genus arguments for compactifications of $E \to D$ constrain $f(E/D)$; the inertia calculation in this paper requires control of the distinct transverse index $e(E/D)$. This is why curvewise finiteness alone does not eliminate a ramified divisor.

3.3 Global and divisorial formulations

Let $\mathfrak{b} \subseteq \overline{A}$ be the radical ideal defining $\partial X = \overline{X} \setminus j(X)$. Because $\overline{X}$ is affine, local cohomology with support in the boundary is

$$H^1_{\partial X}(\overline{X}, \mathcal{O}_{\overline{X}}) = H^1_{\mathfrak{b}}(\overline{A}).$$

Definition 3.10 (Global and divisorial finiteness formulations)**.** For a planar Keller map F, define:

1. The *global boundary formulation*, PlanarBoundaryCoherence, denoted $\mathrm{PBC}(F)$, is the assertion that
$$H^1_{\mathfrak{b}}(\overline{A}) \quad \text{is a finite (equivalently, finitely generated) } \overline{A}\text{-module.}$$

2. The *divisorial differential formulation*, PlanarRamificationRigidity, denoted $\mathrm{PRR}(F)$, is the assertion that
$$\Omega_{A_N/B} \quad \text{has finite length as an } A_N\text{-module.}$$

Here finiteness does not repeat the finiteness already proved in Proposition 3.9. The rings $\overline{A}$ and A_N are finite B-modules because they are integral closures in finite field extensions. Consequently $\overline{X}$ and Z are reduced, irreducible affine schemes of finite type over the algebraically closed field $\mathbb{C}$, hence affine varieties in the usual sense. Ordinary higher cohomology of quasi-coherent sheaves on these affine schemes vanishes (the affine-cohomology theorem, or the vanishing half of Serre's affineness criterion) [The26i]. Neither fact implies that the boundary local cohomology $H^1_{\mathfrak{b}}(\overline{A})$ is finite: local cohomology measures sections with poles along the omitted closed set. Similarly, $\Omega_{A_N/B}$ is automatically a finite A_N-module, because A_N is a finite-type B-algebra, but requiring it to have finite *length* is the stronger assertion that its support has dimension zero.

Proposition 3.11 (The boundary-coherence implication)**.** *Planar boundary coherence gives the implication chain*

$$\mathrm{PBC}(F) \Longrightarrow \partial X = \varnothing \Longrightarrow f_F \textit{ is finite étale} \Longrightarrow F \textit{ is a polynomial automorphism.}$$

Proof. Because $\overline{A}$ is a domain and $j(X)$ contains its generic point, $H^0_{\mathfrak{b}}(\overline{A}) = 0$. The local-cohomology sequence for $j(X) \subseteq \overline{X}$, together with affineness of $\overline{X}$, therefore gives

$$0 \longrightarrow \overline{A} \longrightarrow A \longrightarrow H^1_{\mathfrak{b}}(\overline{A}) \longrightarrow 0.$$

Here $\Gamma(X, \mathcal{O}_X) = A$; this is the standard affine open-complement local-cohomology sequence [The26k]. If the last term is a finite $\overline{A}$-module, then so is A. Thus the open immersion j is finite. Its image is both open and closed; since $\overline{X}$ is integral and $X \neq \varnothing$, it is all of $\overline{X}$. Hence $\partial X = \varnothing$, so $f_F = \bar{\nu}$ is finite. It is already étale by Proposition 3.1, and Proposition 3.2 finishes the proof. □

In particular, $\mathrm{PBC}(F)$ implies $\mathrm{PRR}(F)$: after F is an automorphism one has $A_N = B$, so $\Omega_{A_N/B} = 0$. The converse does not follow by a direct module-theoretic comparison. The ramification-rigidity endgame below nevertheless shows that, for a planar Keller map, $\mathrm{PRR}(F)$ also implies automorphy and hence $\mathrm{PBC}(F)$. The two criteria record different mechanisms: ramification rigidity first eliminates divisorial ramification, whereas boundary coherence eliminates the entire deleted boundary at once.

Let $S_F \subseteq Y$ denote the nonproperness set: the complement of the largest open subset $U \subseteq Y$ over which $f_F^{-1}(U) \to U$ is proper.

Proposition 3.12 (The deleted boundary and properness)**.** *Set-theoretically,*

$$S_F = \bar{\nu}(\partial X).$$

Consequently, the following are equivalent:

$$f_F \text{ is proper}, \qquad S_F = \varnothing, \qquad \partial X = \varnothing.$$

Under any of these conditions, F is a polynomial automorphism.

Proof. The set $\bar{\nu}(\partial X)$ is closed because $\bar{\nu}$ is finite. Over

$$U = Y \setminus \bar{\nu}(\partial X),$$

the open immersion identifies $f_F^{-1}(U)$ with $\bar{\nu}^{-1}(U)$; hence the restriction of f_F is finite and therefore proper. Thus $S_F \subseteq \bar{\nu}(\partial X)$.

Conversely, suppose that $y \in \bar{\nu}(\partial X)$ and that f_F were proper over a neighborhood of y. Since f_F is quasi-finite, it would be finite after shrinking to an affine neighborhood V of y. The normal finite V-scheme $f_F^{-1}(V)$, with function field L, would then be the normalization of V in L. By uniqueness of normalization it would equal $\bar{\nu}^{-1}(V)$, contrary to the existence of a boundary point above y. Therefore $\bar{\nu}(\partial X) \subseteq S_F$, proving the equality and the three equivalences.

If they hold, $f_F = \bar{\nu}$ is finite and étale. Since $Y \cong \mathbb{A}^2_{\mathbb{C}}$, Theorem 3.28 makes this connected cover trivial, so F is a polynomial automorphism. □

Thus the deleted normalization boundary is not auxiliary: it is precisely the geometric source of nonproperness. For a nonsingular polynomial map of the complex plane, a nonempty S_F is known to be a curve with one point at infinity [Cha04]. The inertia calculations below identify which divisorial part of this nonproperness can support the normal closure.

Here dimension two materially sharpens the normalization problem. Both $\overline{X}$ and Z are normal Noetherian surfaces finite over the regular surface Y. A normal Noetherian scheme of dimension at most two is Cohen–Macaulay [The26a]; since the finite maps have zero-dimensional fibers, miracle flatness makes them finite flat [The26l]. Their coordinate rings are therefore finite locally free B-modules, so the trace, different, and ramification divisor belong to the finite-flat surface setting.

Moreover, $j(X)$ is a dense affine open in the separated Noetherian surface $\overline{X}$. Every irreducible component of $\partial X = \overline{X} \setminus j(X)$ consequently has codimension one [The26b]: a nonempty deleted boundary is a union of curves. Purity of the branch locus rules out ramification supported only in codimension two. Thus the remaining planar question is the following concrete one: can a nonzero ramification divisor on Z have every inertia-moving conjugate center on the deleted boundary curves? The coordinate criterion below gives an exact valuation formulation of this question.

Characteristic zero also makes the ramification divisor of the intermediate normalization explicit. Write

$$R_{\bar{\nu}} = \sum_{C \in \overline{X}^{(1)}} (e(C/\bar{\nu}(C)) - 1)C.$$

The different exponent is $e - 1$ at each divisorial valuation, and the étaleness of f_F gives

$$\operatorname{Supp}(R_{\bar{\nu}}) \subseteq \partial X.$$

In canonical-divisor language, $R_{\bar{\nu}}$ is the relative canonical divisor of the finite map. Its vanishing is therefore the finite-cover analogue of crepancy; this is a ramification statement, not a consequence of resolving the isolated singularities of the normal surface $\overline{X}$.

Proposition 3.13 (Boundary divisor-class rigidity)**.** *Let D be an effective Weil divisor on $\overline{X}$ supported on ∂X. If $[D]$ is torsion in $\mathrm{Cl}(\overline{X})$, then $D = 0$.*

Proof. Choose $m > 0$ and $h \in L^\times$ with

$$\mathrm{div}_{\overline{X}}(h) = mD.$$

The divisor on the right is effective. Since $\overline{X}$ is normal, the height-one valuation criterion for regularity gives $h \in \Gamma(\overline{X}, \mathcal{O}_{\overline{X}}) = \overline{A}$. Because D is supported on the deleted boundary, $\mathrm{div}_X(h|_X) = 0$. Since $A = \mathbb{C}[x_1, x_2]$ is a unique factorization domain, this gives

$$h|_X \in A^\times = \mathbb{C}[x_1, x_2]^\times = \mathbb{C}^\times.$$

The inclusion $\overline{A} \subseteq A$ is injective, so h itself is constant. Therefore $mD = 0$, and hence $D = 0$. □

Corollary 3.14 (Different-class criterion for planar rigidity)**.** *If $[R_{\bar{\nu}}]$ is torsion in $\mathrm{Cl}(\overline{X})$, then F is a polynomial automorphism. In particular, this holds if the different is principal or if $\mathrm{Cl}(\overline{X})$ is torsion.*

Proof. Proposition 3.13 gives $R_{\bar{\nu}} = 0$. Purity of the branch locus then makes the finite map $\bar{\nu}$ étale, and finite-étale rigidity of $\mathbb{A}^2_{\mathbb{C}}$ makes it an isomorphism. Thus $L = K$. Proposition 3.1 gives that A is flat over B, so Corollary 2.17 gives $\mathrm{Obs}(F) = 0$. Theorem 2.15 finishes the proof. □

The following specialization recovers the classical Galois case of the Jacobian conjecture [Cam73, Raz79, Wri81] in the boundary language of this section.

Theorem 3.15 (Boundary-theoretic Galois rigidity)**.** *If the marked function-field extension L/K is normal, equivalently Galois, then*

$$L = K \qquad \textit{and} \qquad F \in \mathrm{Aut}_{\mathrm{poly}}(\mathbb{A}^2_{\mathbb{C}}).$$

Proof. Proposition 3.1 shows that L/K is finite and separable, so normality is equivalent to the Galois condition. Let $B_1, \ldots, B_r$ be the prime divisors of Y that are images of components of $R_{\bar{\nu}}$. Since $B = \mathbb{C}[P, Q]$ is a unique factorization domain, choose an irreducible $b_i \in B$ such that

$$B_i = \mathrm{div}_Y(b_i).$$

The Galois group preserves the integral closure $\overline{X}$ and acts transitively on the prime divisors above a fixed B_i, so their ramification indices have a common value e_i [The26h]. Put

$$D_i = \sum_{\substack{C \in \overline{X}^{(1)} \\ \bar{\nu}(C) = B_i}} C.$$

For every $C \mid B_i$, the valuation formula for pullback gives

$$\mathrm{ord}_C(b_i) = e(C/B_i)\,\mathrm{ord}_{B_i}(b_i) = e_i.$$

There are no other height-one zeros of b_i on $\overline{X}$, and hence

$$\mathrm{div}_{\overline{X}}(b_i) = \bar{\nu}^* B_i = e_i D_i.$$

These are the standard Galois valuation and divisor-pullback formulas [The26f, The26p].

The residue characteristic is zero, so the divisorial extensions are tame and the different exponent at every $C \mid B_i$ is $e_i - 1$ [The26q]. Grouping the components of the ramification divisor by their images therefore gives

$$R_{\bar{\nu}} = \sum_{i=1}^{r} (e_i - 1) D_i.$$

Set $m = \prod_{i=1}^{r} e_i$, with the empty product understood as 1, and define

$$h = \prod_{i=1}^{r} b_i^{m(e_i-1)/e_i} \in K^\times \subseteq L^\times.$$

Every exponent is a nonnegative integer, and the preceding pullback formula yields

$$\operatorname{div}_{\overline{X}}(h) = \sum_{i=1}^{r} \frac{m(e_i - 1)}{e_i}\, e_i D_i = m R_{\bar{\nu}}.$$

Thus $[R_{\bar{\nu}}]$ is torsion in $\mathrm{Cl}(\overline{X})$, and Corollary 3.14 gives the result. □

Corollary 3.16 (Quadratic planar rigidity)**.** *No planar Keller map has generic degree $d_F = 2$. Consequently,*

$$d_F \le 2 \quad\Longrightarrow\quad d_F = 1 \quad\Longrightarrow\quad F \in \mathrm{Aut}_{\mathrm{poly}}(\mathbb{A}^2_{\mathbb{C}}).$$

In particular, every nonautomorphic planar Keller map would have

$$d_F \ge 3 \qquad \textit{and} \qquad L/K \ \textit{nonnormal}.$$

Proof. If $d_F = 2$, then the separable quadratic extension L/K is Galois. Theorem 3.15 would give $L = K$, contradicting $[L : K] = 2$. Hence $d_F \le 2$ forces $d_F = 1$; Corollary 2.17 and Theorem 2.15 give automorphy. Finally, Theorem 3.15 rules out normality in every nonautomorphic case. □

Outside this normal/Galois specialization, the divisor-theoretic problem is to determine whether the finite-flat trace and different, together with $\overline{A} \subseteq \mathbb{C}[x_1, x_2]$, force $[R_{\bar{\nu}}]$ to be torsion. The different-class criterion proves the consequence of such torsion; it does not assert it for a general nonnormal extension. Corollary 3.16 shows that only nonnormal extensions of generic degree at least three remain.

It is important that this argument cannot simply be rerun for the always Galois extension N/K. Boundary divisor-class rigidity applies on $\overline{X}$ because it contains the marked affine plane $X \simeq \mathbb{A}^2_{\mathbb{C}}$ as a dense open. If L/K is nonnormal, the Galois normalization Z instead maps finitely to every conjugate intermediate model and need not contain a marked affine-plane open whose deleted boundary supports all of $R_{Z/Y}$. This is precisely why the pulled-back conjugate boundaries and moving-sheet condition are needed.

Let

$$\mathcal{D}_F = \left(Z \xrightarrow{\pi} \overline{X} \xrightarrow{\bar{\nu}} Y,\ j\colon X \hookrightarrow \overline{X}\right), \qquad \partial X = \overline{X} \setminus j(X),$$
$$G = \mathrm{Gal}(N/K), \qquad H = \mathrm{Gal}(N/L).$$

The abstract conjugate centers now admit a direct two-coordinate test. For a representative $g \in G$ of $\omega = D_E g H$, the conjugate affine sheet is generated inside N by

$$A_g = \mathbb{C}[g(x_1), g(x_2)].$$

The two derivations of Proposition 3.4, restricted to K, extend uniquely across the finite separable extension N/K. Indeed, if $z \in N$ has separable minimal polynomial $f \in K[X]$, any extension is forced by

$$\widetilde{\partial}(z) = -\frac{(\partial f)(z)}{f'(z)}, \qquad f'(z) \neq 0.$$

For $g \in G$, both $g\widetilde{\partial}g^{-1}$ and $\widetilde{\partial}$ extend the same derivation of K, so uniqueness makes them equal. Consequently

$$\widetilde{\partial}(g(a)) = g(\partial(a)) \in g(A) \qquad (a \in A),$$

and the extended inverse-Jacobian frame preserves every A_g. No preservation of the integral order $T = A_N$ is asserted.

Proposition 3.17 (Planar coordinate test for a conjugate center). *For $E \in Z^{(1)}$ and $\omega = D_E gH \in \mathcal{Q}_E$,*

$$\mathrm{cent}(E,\omega) \in j(X) \quad\Longleftrightarrow\quad w_E(g(x_1)) \geq 0 \text{ and } w_E(g(x_2)) \geq 0.$$

Consequently,

$$\mathrm{cent}(E,\omega) \in \partial X \quad\Longleftrightarrow\quad \min\{w_E(g(x_1)), w_E(g(x_2))\} < 0.$$

Proof. By definition, $u_\omega = (g^{-1}w_E)|_L$, so

$$u_\omega(x_i) = w_E(g(x_i)) \qquad (i = 1, 2).$$

The center of u_ω lies on the affine open $X = \operatorname{Spec}\mathbb{C}[x_1, x_2]$ exactly when its valuation ring contains A. Since A is generated by x_1, x_2, this is equivalent to the two displayed nonnegativity conditions. Negating them gives the boundary criterion. □

The early secant estimate has a precise interpretation on these conjugate generic sheets. For $g \in G$, let

$$\delta_{F,g} := \delta_F(x_1, x_2, g(x_1), g(x_2)) \in N, \qquad q_{F,g} := 1 - c^{-1}\delta_{F,g}.$$

Proposition 3.18 (Secant evaluation on conjugate sheets). *For every $g \in G$,*

$$\delta_{F,g}(x_i - g(x_i)) = 0 \qquad (i = 1, 2).$$

Consequently,

$$\delta_{F,g} = \begin{cases} c, & g \in H, \\ 0, & g \notin H, \end{cases} \qquad q_{F,g} = \begin{cases} 0, & g \in H, \\ 1, & g \notin H. \end{cases}$$

Thus the planar secant projector is exactly the characteristic function of the nonmarked generic conjugate sheets.

Proof. The marked embedding $L \hookrightarrow N$ and its g-conjugate agree on K, so together they define an evaluation map from the generic collision algebra $L \otimes_K L$ to N. Evaluating Proposition 3.5 under this map gives the first display. If $g \in H = \mathrm{Gal}(N/L)$, then $g(x_i) = x_i$ for both coordinates, and diagonal evaluation gives $\delta_{F,g} = c$. If $g \notin H$, then some x_i is moved, since $L = \mathbb{C}(x_1, x_2)$. The corresponding factor $x_i - g(x_i)$ is nonzero in the field N, so the first display forces $\delta_{F,g} = 0$. The formulas for $q_{F,g}$ follow. □

The complementary idempotent $e_F = 1 - q_F$ gives an explicit trace-theoretic bridge. Under the standard finite Galois decomposition

$$N \otimes_K N \xrightarrow{\sim} \prod_{\sigma \in G} N, \qquad a \otimes b \longmapsto \big(a\,\sigma(b)\big)_\sigma,$$

the generic base change of e_F is the characteristic function of H. Let $e_g = (g \otimes g)e_F$; then e_g is the characteristic function of gHg^{-1}. Proposition 2.8 gives

$$e_N := \prod_{g \in G} e_g = \mathbf{1}_{\mathrm{core}_G(H)} = \mathbf{1}_{\{1\}}.$$

Finally, the trace pairing gives a canonical K-linear isomorphism

$$\Theta_N : N \otimes_K N \xrightarrow{\sim} \mathrm{End}_K(N), \qquad \Theta_N(a \otimes b)(z) = a\,\mathrm{Tr}_{N/K}(bz),$$

and $\Theta_N(e_N) = \mathrm{id}_N$. Thus the product of the conjugate planar secant idempotents is the trace coevaluation tensor for N/K. Notice that Θ_N is a linear, not a ring, isomorphism. For the finite locally free order $T = A_N$, the integral coevaluation tensor lies in $T \otimes_B \mathrm{Hom}_B(T, B)$; its generic fiber is e_N after the trace identification. This identifies the finite trace-dual lattice canonically, but does not yet bound poles of moving-sheet principal parts.

This follows the exponent-one secant bound completely into the normal closure, but also locates its limit. On a nonmarked sheet $\delta_{F,g}$ is the zero function; it is therefore a generic-sheet projector, not a nonzero boundary parameter whose fixed power bounds pole order. Any use of the secant matrix for uniform boundary control therefore needs a further transport statement, using more than the determinant alone. That possible mechanism is isolated after the completed endgame.

The generic idempotents and the conjugate affine opens play different roles. The former separate components of the generic collision algebra, whereas $U_g \subset Z$ tests whether the corresponding conjugate center remains integral. At a ramified divisor those generic components can meet in an integral model, so no extension of e_g as an integral idempotent is asserted. The research mechanism below instead attaches to this generic sheet isolation a fixed–moving boundary contribution supported on

$$V(\mathfrak{a}_C + \mathfrak{j}_C^{\mathrm{mov}}).$$

For $E \in \mathrm{Ram}^{(1)}(Z/Y)$ and $\omega \in \mathcal{Q}_E$, Proposition 2.11 gives

$$e\big(\mathrm{cent}(E, \omega)/\nu(E)\big) = i_E(\omega).$$

Since $X \to Y$ is étale by Proposition 3.1, Proposition 2.12 gives

$$i_E(\omega) > 1 \quad \Longrightarrow \quad \mathrm{cent}(E, \omega) \in \partial X.$$

Thus Definition 2.13 specializes to

$$\mathcal{H}(\mathcal{D}_F) = \left\{ E \in \mathrm{Ram}^{(1)}(Z/Y) : \begin{array}{l} \exists \omega \in \mathcal{Q}_E,\ i_E(\omega) > 1, \\ \forall \omega \in \mathcal{Q}_E, \quad i_E(\omega) > 1 \Longrightarrow \mathrm{cent}(E, \omega) \in \partial X \end{array} \right\}.$$

Proposition 3.17 makes this locus fully explicit:

$$\mathcal{H}(\mathcal{D}_F) = \left\{ E \in \mathrm{Ram}^{(1)}(Z/Y) : \begin{array}{l} \forall g \in G, \\ [I_E : I_E \cap gHg^{-1}] > 1 \Longrightarrow \min\{w_E(g(x_1)), w_E(g(x_2))\} < 0 \end{array} \right\}.$$

The existence of at least one g with nontrivial relative index is automatic from $I_E \neq 1$ and $\operatorname{core}_G(H) = 1$. Thus hidden inertia is not an additional abstract object in the planar proof: it is precisely the possibility that every inertia-moving conjugate coordinate pair acquires a pole at the deleted boundary.

Definition 3.19 (Planar conjugate coverage)**.** Write $\mathrm{Cov}_2(F)$ for the following condition: every $E \in \mathrm{Ram}^{(1)}(Z/Y)$ admits a $g \in G$ such that

$$[I_E : I_E \cap gHg^{-1}] > 1, \qquad w_E(g(x_1)) \geq 0, \qquad w_E(g(x_2)) \geq 0.$$

Thus some conjugate sheet moved by inertia still has an affine center.

We now construct the canonical boundaries directly on the common Galois normalization. For $g \in G$, set

$$\overline{A}_g = g(\overline{A}), \qquad \overline{X}_g = \operatorname{Spec} \overline{A}_g, \qquad X_g = \operatorname{Spec} A_g.$$

The conjugate of j is an open immersion $X_g \hookrightarrow \overline{X}_g$, and the inclusion $\overline{A}_g \subseteq A_N$ gives a finite map $\pi_g \colon Z \to \overline{X}_g$. Let

$$U_g := \pi_g^{-1}(X_g) \subseteq Z, \qquad D_g := Z \setminus U_g, \qquad \mathfrak{j}_g := I_{A_N}(D_g) = \bigcap_{\mathfrak{p} \in D_g} \mathfrak{p}.$$

Thus $\mathfrak{j}_g$ is canonically the radical vanishing ideal of the pulled-back deleted boundary and $D_g = V(\mathfrak{j}_g)$. The construction depends only on gH, since right multiplication by H does not change the conjugate embedding of L.

Proposition 3.20 (Pulled-back boundary dictionary)**.** *For every height-one point E of Z and every $g \in G$,*

$$\eta_E \in U_g \iff \mathfrak{j}_g \not\subseteq \mathfrak{p}_E \iff w_E(g(x_1)) \geq 0 \quad \textit{and} \quad w_E(g(x_2)) \geq 0.$$

Equivalently,

$$\mathfrak{j}_g \subseteq \mathfrak{p}_E \iff \eta_E \in D_g \iff \min\{w_E(g(x_1)), w_E(g(x_2))\} < 0.$$

Proof. Because $D_g = V(\mathfrak{j}_g)$, membership $\eta_E \in D_g$ is exactly the containment $\mathfrak{j}_g \subseteq \mathfrak{p}_E$. The point η_E belongs to U_g precisely when its image under π_g is centered on $X_g = \operatorname{Spec} \mathbb{C}[g(x_1), g(x_2)]$. The valuation criterion for a center on an affine model says exactly that both conjugate coordinates have nonnegative valuation. Taking complements gives the second display. $\square$

For a nontrivial subgroup $I \leq G$, define its moving-boundary ideal by

$$\mathfrak{j}_I^{\mathrm{mov}} := \sum_{\substack{g \in G \\ I \not\subseteq gHg^{-1}}} \mathfrak{j}_g.$$

The sum is taken in the common Galois-normalization ring A_N; this is why the Galois closure enters only at this final specialization.

Proposition 3.21 (Moving-sheet coverage dictionary)**.** *For $E \in \mathrm{Ram}^{(1)}(Z/Y)$, one has*

$$\exists g \in G, \quad I_E \not\subseteq gHg^{-1}, \quad \eta_E \in U_g \iff \mathfrak{j}_{I_E}^{\mathrm{mov}} \not\subseteq \mathfrak{p}_E.$$

Thus eliminating E is exactly the failure of its moving-boundary ideal to be contained in its height-one prime.

Proof. A prime contains a sum of ideals exactly when it contains every summand. Apply Proposition 3.20 to each inertia-moving g. $\square$

The same common ring gives a finite-group formulation which does not first choose a ramified divisor. For a subgroup $C \le G$, let

$$\mathfrak{a}_C := \big(\sigma(t) - t : \sigma \in C,\ t \in A_N\big) \subseteq A_N.$$

This is the scheme-theoretic fixed-locus ideal for the action of C on Z. If $\mathfrak{p}_E \in \operatorname{Spec} A_N$, then

$$\mathfrak{a}_C \subseteq \mathfrak{p}_E \quad \Longleftrightarrow \quad C \le I_E.$$

Indeed, the containment says that every element of C stabilizes $\mathfrak{p}_E$ and acts trivially on its residue field, which is the definition of inertia at E.

Definition 3.22 (Planar boundary separation)**.** Write $\mathrm{Sep}_2(F)$ for the following condition: for every prime-order subgroup $C \le G$,

$$\operatorname{Spec}^{(1)}(A_N) \cap V(\mathfrak{a}_C + \mathfrak{j}_C^{\mathrm{mov}}) = \varnothing.$$

Equivalently,

$$\forall \mathfrak{p} \in \operatorname{Spec}^{(1)}(A_N), \qquad \mathfrak{a}_C \subseteq \mathfrak{p} \implies \mathfrak{j}_C^{\mathrm{mov}} \not\subseteq \mathfrak{p}.$$

In codimension language, every fixed–moving boundary locus $V(\mathfrak{a}_C + \mathfrak{j}_C^{\mathrm{mov}})$ has codimension at least two in the normal surface Z. No assertion is made about isolated fixed boundary points.

Proposition 3.23 (Boundary separation is the divisorial criterion)**.** *For a planar Keller map,*

$$\mathrm{Sep}_2(F) \quad \Longleftrightarrow \quad \mathrm{Ram}^{(1)}(Z/Y) = \varnothing.$$

Proof. Assume boundary separation and suppose that $E \in \mathrm{Ram}^{(1)}(Z/Y)$. Choose a subgroup $C \le I_E$ of prime order; such a subgroup exists by Cauchy's theorem for the nontrivial finite group I_E. Then $\mathfrak{a}_C \subseteq \mathfrak{p}_E$. If $C \not\subseteq gHg^{-1}$, then $I_E \not\subseteq gHg^{-1}$, so the relative inertia index on the sheet represented by gH is nontrivial. Proposition 2.12 places its center in the deleted boundary, and hence $\mathfrak{j}_g \subseteq \mathfrak{p}_E$. This holds for every sheet moved by C, so

$$\mathfrak{a}_C + \mathfrak{j}_C^{\mathrm{mov}} \subseteq \mathfrak{p}_E,$$

contrary to boundary separation.

Conversely, if boundary separation fails, there are a prime-order $C \le G$ and a height-one prime $\mathfrak{p}_E$ containing $\mathfrak{a}_C + \mathfrak{j}_C^{\mathrm{mov}}$. The fixed-locus containment gives $C \le I_E$. Since $C \ne 1$, the point E is ramified, contradicting $\mathrm{Ram}^{(1)}(Z/Y) = \varnothing$. $\square$

Proposition 3.24 (Coverage eliminates a ramified divisor)**.** *Let $E \in \mathrm{Ram}^{(1)}(Z/Y)$. If there is a $g \in G$ such that*

$$I_E \not\subseteq gHg^{-1} \qquad \text{and} \qquad \eta_E \in U_g,$$

then a contradiction follows. Consequently,

$$\mathrm{Cov}_2(F) \quad \Longrightarrow \quad \mathrm{Ram}^{(1)}(Z/Y) = \varnothing.$$

Proof. The subgroup condition gives

$$i_E(gH) = [I_E : I_E \cap gHg^{-1}] > 1.$$

On the other hand, $\eta_E \in U_g$ places the conjugate center on the affine sheet X_g. The map $X_g \to Y$ is the Galois conjugate of the planar Keller map and is therefore étale. Its local ramification index at that center is one. Hence $1 < i_E(gH) = 1$, a contradiction. □

This is the point at which a height-one divisor is eliminated. Purity is not used in Proposition 3.24; it enters only after the proposition has been applied to every ramified height-one point.

3.4 Moving-sheet criteria for planar ramification rigidity

Corollary 3.25 (Planar hidden-inertia identification)**.**

$$\mathcal{H}(\mathcal{D}_F) = \mathrm{Ram}^{(1)}(Z/Y).$$

In particular,

$$\mathcal{H}(\mathcal{D}_F) = \varnothing \quad \Longleftrightarrow \quad \mathrm{Ram}^{(1)}(Z/Y) = \varnothing.$$

These conditions are also equivalent to planar conjugate coverage.

Proof. The equality is Proposition 2.14, applied to the planar finite-cover diagram. Conjugate coverage implies an empty ramification locus by Proposition 3.24, and hence an empty hidden-inertia locus by the displayed equality. Conversely, if the ramification locus is empty, conjugate coverage holds vacuously. □

Proposition 3.26 (The ramification-rigidity dictionary)**.** *For a planar Keller map,*

$$\mathrm{PRR}(F) \quad \Longleftrightarrow \quad \mathrm{Ram}^{(1)}(Z/Y) = \varnothing \quad \Longleftrightarrow \quad \mathrm{Cov}_2(F) \quad \Longleftrightarrow \quad \mathrm{Sep}_2(F).$$

Thus moving-sheet coverage and boundary separation are two exact criteria for PlanarRamificationRigidity, not additional global hypotheses.

Proof. The A_N-module $\Omega_{A_N/B}$ is finite, and its generic localization vanishes because N/K is separable. Since A_N is a two-dimensional affine $\mathbb{C}$-domain, every positive-dimensional closed subset has a height-one generic point. Thus this finite module has finite length exactly when it has no height-one point in its support. At a height-one point E, localizing B first at the image of E and then at $\mathfrak{p}_E$ gives the relevant finite local extension; the source local ring is a discrete valuation ring. In characteristic zero its residue extension is separable, so

$$\Omega_{(A_N)_{\mathfrak{p}_E}/B_{\mathfrak{p}_E \cap B}} = 0 \quad \Longleftrightarrow \quad I_E = 1.$$

Indeed, vanishing of the localized differential module is the finite-type unramifiedness criterion, and for a finite extension of discrete valuation rings in characteristic zero unramifiedness is equivalent to trivial inertia [The26j, The26f, The26p]. Consequently finite length is equivalent to the absence of divisorial ramification. Proposition 3.24 and vacuity in the unramified case give the equivalence with coverage; the equivalence with separation is Proposition 3.23. □

Equivalently, the premise $\mathcal{H}(\mathcal{D}_F) = \varnothing$ says that if a ramified divisor existed, some inertia-moving conjugate pair $g(x_1), g(x_2)$ would have to remain integral and hence centered on an affine sheet. Planar étaleness says that the same pair must lie at the boundary. This is the boundary contradiction used below; the premise supplies its exclusion, while the preceding results identify exactly what is being excluded.

3.5 The divisorial endgame

Theorem 3.27 (Purity of the branch locus, specialized)**.** *Let $Z \to \mathbb{A}^2_{\mathbb{C}}$ be a finite dominant generically separable morphism with Z integral and normal. If $\mathrm{Ram}^{(1)}(Z/\mathbb{A}^2_{\mathbb{C}}) = \varnothing$, then the morphism is étale.*

Proof. At every height-one point of Z, the absence of divisorial ramification gives ramification index one. The corresponding residue field extension is finite and, in characteristic zero, separable. Hence the morphism is unramified at every height-one point. Purity of the branch locus now implies that it is étale everywhere [The26o]. □

Theorem 3.28 (Finite-étale rigidity of the affine plane)**.** *Every connected finite étale cover of $\mathbb{A}^2_{\mathbb{C}}$ is trivial [GR71, Exposé XII, Théorème 5.1].*

Corollary 3.29 (Triviality of the planar normal closure)**.** *If*

$$\mathrm{Ram}^{(1)}(Z/Y) = \varnothing,$$

then

$$N = K \qquad \textit{and hence} \qquad L = K.$$

Numerically,

$$[L : K] = [N : L] = [N : K] = 1.$$

Proof. By Proposition 3.1, $Y \cong \mathbb{A}^2_{\mathbb{C}}$. Since N/K is finite Galois, $\nu \colon Z \to Y$ is generically separable; hence Theorem 3.27 makes it finite étale. The inclusion

$$A_N \subset N \text{ is a domain,} \qquad \text{so} \qquad Z = \operatorname{Spec} A_N \text{ is connected.}$$

Theorem 3.28 therefore makes $\nu \colon Z \to Y$ an isomorphism. Taking function fields gives $N = K$. Finally, $K \subseteq L \subseteq N$ implies $L = K$. □

Corollary 3.30 (The divisorial endgame)**.** *The absence of divisorial ramification gives the implication chain*

$$\mathrm{Ram}^{(1)}(Z/Y) = \varnothing \Longrightarrow Z \longrightarrow Y \textit{ is finite étale} \Longrightarrow N = K \Longrightarrow L = K \Longrightarrow q_F = 0 \Longrightarrow I_R = I_\Delta.$$

Proof. Purity, Theorem 3.27, makes $Z \to Y$ finite étale, and Corollary 3.29 gives $N = K$ and $L = K$. Thus $d_F = 1$. Flatness of A over B and Corollary 2.17 give $\mathrm{Obs}(F) = 0$. The abstract off-diagonal projector and collision criteria then give $q_F = 0$ and $I_R = I_\Delta$. □

By Proposition 3.26, any of $\mathrm{PRR}(F)$, $\mathrm{Sep}_2(F)$, or $\mathrm{Cov}_2(F)$ supplies the first condition in this single endgame. They are respectively differential, ideal-theoretic, and valuation-theoretic descriptions of its codimension-one premise, rather than three additional routes.

3.6 Vanishing of the off-diagonal factor

The next theorem records all named formulations as exact translations of the preceding divisorial endgame.

Theorem 3.31 (Equivalent boundary criteria for planar vanishing)**.** *For every polynomial map* $F = (P, Q) \colon \mathbb{A}^2_{\mathbb{C}} \to \mathbb{A}^2_{\mathbb{C}}$ *satisfying* $\det JF = c \in \mathbb{C}^\times$*, the following conditions are equivalent:*

$$\begin{aligned} \mathrm{PBC}(F) \quad &\Longleftrightarrow \quad \mathrm{PRR}(F) \quad \Longleftrightarrow \quad \mathrm{Sep}_2(F) \quad \Longleftrightarrow \quad \mathrm{Cov}_2(F) \\ &\Longleftrightarrow \quad \mathcal{H}(\mathcal{D}_F) = \varnothing \quad \Longleftrightarrow \quad \mathrm{Ram}^{(1)}(Z/Y) = \varnothing, \end{aligned}$$

and

$$\mathrm{Ram}^{(1)}(Z/Y) = \varnothing \quad \Longleftrightarrow \quad d_F = 1 \quad \Longleftrightarrow \quad \mathrm{Obs}(F) = 0,$$
$$\mathrm{Obs}(F) = 0 \quad \Longleftrightarrow \quad q_F = 0 \quad \Longleftrightarrow \quad R_F^\circ = \varnothing \quad \Longleftrightarrow \quad I_R = I_\Delta.$$

Proof. Proposition 3.11 shows that $\mathrm{PBC}(F)$ implies that F is an automorphism. Then $B = A$, $K = L = N$, $\partial X = \varnothing$, and $\Omega_{A_N/B} = 0$, so $\mathrm{PRR}(F)$ holds. Conversely, the argument below shows that $\mathrm{PRR}(F)$ implies automorphy; for an automorphism $\partial X = \varnothing$, and hence $H^1_{\partial X}(\overline{X}, \mathcal{O}_{\overline{X}}) = 0$. Thus $\mathrm{PBC}(F)$ holds as well.

Proposition 3.26 identifies $\mathrm{PRR}(F)$ and $\mathrm{Sep}_2(F)$ with the absence of divisorial ramification. Proposition 3.24 shows directly that $\mathrm{Cov}_2(F)$ implies $\mathrm{Ram}^{(1)}(Z/Y) = \varnothing$; conversely, if the ramification locus is empty, coverage holds vacuously. Corollary 3.25 identifies these conditions with $\mathcal{H}(\mathcal{D}_F) = \varnothing$.

Under these conditions, Corollary 3.29 gives $N = L = K$, hence $d_F = 1$, while Proposition 3.1 makes A flat over B. Corollary 2.17 therefore gives $\mathrm{Obs}(F) = 0$. Conversely, if $d_F = 1$, then $L = K$; flatness and Corollary 2.17 again give $\mathrm{Obs}(F) = 0$. If $\mathrm{Obs}(F) = 0$, then $I_R = I_\Delta$, so Theorem 2.15 makes F a polynomial automorphism. Hence $B = A$, $K = L = N$, and $\mathrm{Ram}^{(1)}(Z/Y) = \varnothing$.

Corollary 2.4 and Theorem 2.5 state

$$R_F^\circ = \varnothing \quad \Longleftrightarrow \quad C_F^\circ = 0 \quad \Longleftrightarrow \quad \mathrm{Obs}(F) = 0 \quad \Longleftrightarrow \quad I_R = I_\Delta,$$
$$\mathrm{Obs}(F) = 0 \quad \Longleftrightarrow \quad q_F = 0.$$

This proves the remaining equivalences. □

Corollary 3.32 (Boundary witness for failure of coverage)**.** *The equivalent conditions of Theorem 3.31 fail if and only if there is an* $E \in \mathrm{Ram}^{(1)}(Z/Y)$ *such that*

$$\forall g \in G, \qquad [I_E : I_E \cap gHg^{-1}] > 1 \quad \Longrightarrow \quad \min\{w_E(g(x_1)), w_E(g(x_2))\} < 0.$$

For such an E, at least one inertia-moving g exists. Thus failure of planar collision vanishing is witnessed by a ramified divisor whose inertia-moving conjugate centers all lie on the deleted boundary. By Theorem 3.15 and Corollary 3.16, its marked extension L/K is nonnormal and its generic degree satisfies $d_F \geq 3$.

Proof. If the equivalent conditions fail, then conjugate coverage fails. Negating Definition 3.19 supplies an $E \in \mathrm{Ram}^{(1)}(Z/Y)$ for which every g of nontrivial relative index fails at least one valuation inequality, giving the displayed condition. Conversely, the displayed condition is incompatible with a coverage witness for E, so conjugate coverage fails. Core-freeness guarantees the existence of at least one g of nontrivial relative index. □

Corollary 3.33 (Exact boundary forms of the planar Jacobian conjecture)**.** *The following universal statements are equivalent:*

1. *The classical complex two-dimensional Jacobian conjecture holds: every polynomial self-map of $\mathbb{A}^2_{\mathbb{C}}$ with constant nonzero Jacobian determinant is a polynomial automorphism.*
2. *Every planar Keller map satisfies* $\mathrm{PBC}(F)$*.*
3. *Every planar Keller map satisfies* $\mathrm{PRR}(F)$*.*
4. *Every planar Keller map satisfies* $\mathrm{Sep}_2(F)$*.*
5. *Every planar Keller map satisfies* $\mathrm{Cov}_2(F)$*.*

Equivalently, condition (4) *says that for every planar Keller map and every prime-order subgroup* $C \leq \mathrm{Gal}(N/K)$*,*

$$\operatorname{codim}_Z V(\mathfrak{a}_C + \mathfrak{j}_C^{\mathrm{mov}}) \geq 2.$$

Proof. If $JC(2)$ holds, every planar Keller map is an automorphism. Then $B = A$, $K = L = N$, $\overline{X} = X$, and $\partial X = \varnothing$, so both $H^1_{\partial X}(\overline{X}, \mathcal{O}_{\overline{X}})$ and $\Omega_{A_N/B}$ vanish. Thus (1) implies (2) and (3). Conversely, Proposition 3.11 shows that (2) implies (1). Theorem 3.31 identifies, for every planar Keller map,

$$\mathrm{PRR}(F) \iff \mathrm{Sep}_2(F) \iff \mathrm{Cov}_2(F) \iff \mathrm{Obs}(F) = 0.$$

Corollary 2.16 identifies universal obstruction vanishing with classical $JC(2)$, proving the remaining equivalences. The prime-order ideal formulation is Proposition 3.23. □

The planar section therefore gives an exact codimension-one reformulation. Here "divisorially coprime" means that $\mathfrak{a}_C$ and $\mathfrak{j}_C^{\mathrm{mov}}$ are contained in no common height-one prime; it does not mean that their sum is the unit ideal. Because Z is an integral surface, the relevant locus has only two possibilities: it has a codimension-one component, which is precisely the bad ramified-divisor case, or it has codimension at least two, in which case only isolated closed points may remain (or the locus is empty). There is no nonempty codimension-three case on Z, and purity is why these possible codimension-two points need not be eliminated separately.

In the good case, conjugate coverage excludes complete boundary-supported inertia orbits and the collision relation collapses to the diagonal. In the bad case, Corollary 3.32 produces exactly such a divisorial boundary witness. PlanarBoundaryCoherence records the equivalent global-boundary formulation that removes the entire normalization boundary at once.

3.7 Research mechanisms and status

The preceding equivalence theorem is the completed logical dictionary. The following mechanisms ask how its divisorial premise might be established from the special polynomial-plane data; they are not additional steps in the endgame.

3.7.1 Two statements and the missing secant–trace morphism

Put $T := A_N$. For every prime-order subgroup $C \leq G$, write

$$\mathfrak{J}_C := \mathfrak{a}_C + \mathfrak{j}_C^{\mathrm{mov}}, \qquad \mathcal{P}_C^{\mathrm{fm}} := H^1_{\mathfrak{J}_C}(T).$$

The ideal $\mathfrak{J}_C$ cuts out the scheme-theoretic intersection of the C-fixed locus with the boundaries of all sheets moved by C. Thus $\mathfrak{J}_C \subseteq \mathfrak{p}_E$ is exactly the fixed–moving height-one configuration isolated by boundary separation.

Proposition 3.34 (I. The fixed–moving pole tower)**.** *For every height-one prime $\mathfrak{p} \subset T$,*

$$\left(\mathcal{P}_C^{\mathrm{fm}}\right)_{\mathfrak{p}} \cong \begin{cases} 0, & \mathfrak{J}_C \nsubseteq \mathfrak{p}, \\ \mathrm{Frac}(T_{\mathfrak{p}})/T_{\mathfrak{p}}, & \mathfrak{J}_C \subseteq \mathfrak{p}. \end{cases}$$

In the second case this localization has principal parts of arbitrarily large pole order and is not a finite $T_{\mathfrak{p}}$-module. Moreover,

$$\mathrm{Sep}_2(F) \quad \Longleftrightarrow \quad \mathcal{P}_C^{\mathrm{fm}} = 0 \quad \textit{for every prime-order } C \leq G.$$

Proof. The action of G on T is faithful, so $\mathfrak{J}_C \neq 0$ for $C \neq 1$. Local cohomology commutes with localization. If $\mathfrak{J}_C \nsubseteq \mathfrak{p}$, its extension to $T_{\mathfrak{p}}$ is the unit ideal and the localization vanishes. If $\mathfrak{J}_C \subseteq \mathfrak{p}$, normality makes $T_{\mathfrak{p}}$ a DVR, and the one-element Čech complex gives

$$H^1_{\mathfrak{J}_C T_{\mathfrak{p}}}(T_{\mathfrak{p}}) \cong \mathrm{Frac}(T_{\mathfrak{p}})/T_{\mathfrak{p}}.$$

The classes of π^{-m}, for a uniformizer π and $m \geq 1$, have unbounded pole order, proving nonfiniteness.

If boundary separation holds, no height-one prime contains $\mathfrak{J}_C$. The normal Noetherian surface T is S_2, so $\mathfrak{J}_C$ has grade at least two and $H^1_{\mathfrak{J}_C}(T) = 0$. Conversely, failure of boundary separation gives a height-one prime containing $\mathfrak{J}_C$, and the displayed localization shows that $\mathcal{P}_C^{\mathrm{fm}} \neq 0$. □

Proposition 3.35 (II. The trace-dual bounded stage)**.** *Define*

$$T^\dagger := \{z \in N : \mathrm{Tr}_{N/K}(zT) \subseteq B\}, \qquad Q_{\mathrm{tr}} := T^\dagger/T.$$

Then the trace pairing gives

$$T^\dagger \cong \mathrm{Hom}_B(T, B),$$

and Q_{tr} is a finite T-module. If $\mathfrak{p}_E \subset T$ has ramification index e_E and uniformizer π_E, then

$$(Q_{\mathrm{tr}})_{\mathfrak{p}_E} \cong \pi_E^{-(e_E-1)} T_{\mathfrak{p}_E}/T_{\mathfrak{p}_E}.$$

Thus the trace-dual quotient contains only one bounded pole stage. Equivalently, for

$$\omega_Z := \mathrm{Hom}_B(T, \omega_Y), \qquad \eta_F := c^{-1} dP \wedge dQ,$$

finite duality gives

$$Q_F^\omega := \omega_Z/T\eta_F \cong Q_{\mathrm{tr}}.$$

Proof. The finite locally free B-algebra T and generic separability of N/K identify $T^\dagger$ with $\mathrm{Hom}_B(T, B)$ through the trace pairing. Since B is normal, traces of elements integral over B lie in B; hence $T \subseteq T^\dagger$, and the quotient is finite. After height-one localization, characteristic-zero ramification is tame and the different exponent is $e_E - 1$, giving the displayed inverse-different formula [The26c, The26d].

Trivializing ω_Y by $dP \wedge dQ$ identifies ω_Z with the trace dual. The Keller identity identifies $T\eta_F$ with the submodule $T \subseteq T^\dagger$, and therefore identifies the two quotients [The26e]. □

The trace transporter. Fix a prime-order subgroup $C \le G$, and put

$$U_C := \operatorname{Spec} T \setminus V(\mathfrak{J}_C), \qquad R_C := \Gamma(U_C, \mathcal{O}_{U_C}) \subseteq N.$$

Faithfulness of the G-action gives $\mathfrak{J}_C \neq 0$, so U_C is dense and its ring of regular functions embeds in $\operatorname{Frac}(T) = N$. Since T is a domain, $H^0_{\mathfrak{J}_C}(T) = 0$, and the affine open-complement sequence gives

$$0 \longrightarrow T \longrightarrow R_C \longrightarrow H^1_{\mathfrak{J}_C}(T) \longrightarrow 0, \qquad \mathcal{P}^{\mathrm{fm}}_C \cong R_C/T\text{[The26k]}.$$

Thus $T \subseteq T^\dagger \subseteq N$ places both $\mathcal{P}^{\mathrm{fm}}_C$ and Q_{tr} in N/T. Define the trace-transporter ideal

$$\mathfrak{c}^{\mathrm{tr}}_C := \{s \in T : sR_C \subseteq T^\dagger\}.$$

For $s \in T$, the following conditions are equivalent:

(i) $s \in \mathfrak{c}^{\mathrm{tr}}_C$;

(ii)
$$\operatorname{Tr}_{N/K}(srt) \in B \qquad (r \in R_C,\ t \in T);$$

(iii) multiplication by s on N/T restricts to the unique T-linear morphism

$$\lambda_{C,s} \colon \mathcal{P}^{\mathrm{fm}}_C \longrightarrow Q_{\mathrm{tr}}, \qquad [r] \longmapsto [sr].$$

Indeed, the first two conditions are equivalent by the definition of $T^\dagger$. Under condition (i), one has $sR_C \subseteq T^\dagger$, while $sT \subseteq T$; hence multiplication by s descends to the quotient map in (iii). This is the exact landing morphism; it is not required to be injective.

The secant–frame denominator candidate. For $g \in G$, evaluate the full secant matrix and its adjugate on the g-conjugate sheet:

$$M_{F,g} := M_F(x, g(x)) \in M_2(N), \qquad A_{F,g} := \operatorname{adj}(M_{F,g}).$$

The extended Keller frame of Proposition 3.4 consists of derivations $\widetilde{\partial}_P, \widetilde{\partial}_Q$ on N. They commute: their commutator vanishes on $K = \mathbb{C}(P,Q)$, and uniqueness of extension across the finite separable extension N/K makes it zero on N. For $\alpha = (a,b) \in \mathbf{N}^2$, set $D^\alpha = \widetilde{\partial}_P^a \widetilde{\partial}_Q^b$, and, for $r \ge 0$, let

$$\Xi_{g,r} := \{D^\alpha (M_{F,g})_{ij}, D^\alpha (A_{F,g})_{ij} : 1 \le i,j \le 2,\ |\alpha| \le r\} \subset N.$$

Put

$$\Lambda^{\mathrm{sf}}_{C,r} := T + \sum_{\substack{g \in G \\ C \not\subseteq gHg^{-1}}} \sum_{\xi \in \Xi_{g,r}} T\xi \subseteq N, \qquad \mathfrak{d}^{\mathrm{sf}}_{C,r} := \{s \in T : s\Lambda^{\mathrm{sf}}_{C,r} \subseteq T\}.$$

Indexing over all $g \in G$ avoids any choice of coset representatives; the family is still finite. The case $r = 1$ is the first-jet candidate obtained from the secant matrix, its adjugate, and the inverse-Jacobian frame.

Proposition 3.36 (Nonzero secant–frame denominator)**.** *For every C and every $r \ge 0$,*

$$\mathfrak{d}^{\mathrm{sf}}_{C,r} \neq 0.$$

Moreover, every nonzero element of this ideal remains nonzero in $T_\mathfrak{p}$ for every prime $\mathfrak{p} \subset T$.

Proof. The set of generators in $\Lambda^{\mathrm{sf}}_{C,r}$ is finite, and each belongs to $N = \operatorname{Frac}(T)$. Clearing their finitely many denominators produces $0 \neq s \in T$ with $s\Lambda^{\mathrm{sf}}_{C,r} \subseteq T$. The localization assertion follows because T is a domain. ☐

The monogenic trace-comparison order. The finite separable extension N/K has a primitive element [The26n]. Clearing one B-denominator gives an integral primitive generator $\alpha_{\mathrm{sec}} \in T$ with $N = K(\alpha_{\mathrm{sec}})$. Let

$$R_{\mathrm{sec}} := B[\alpha_{\mathrm{sec}}] \subseteq T, \qquad f_{\mathrm{sec}}(Z) = Z^d + \gamma_{d-1}Z^{d-1} + \cdots + \gamma_0 \in B[Z]$$

be its monic minimal polynomial, where $d = [N : K]$ and $\gamma_d := 1$. The coefficients lie in B, because they lie in K, are integral over B, and B is normal. Consequently

$$R_{\mathrm{sec}} \cong B[Z]/(f_{\mathrm{sec}}), \qquad R_{\mathrm{sec}} = \bigoplus_{j=0}^{d-1} B\alpha_{\mathrm{sec}}^j, \qquad \mathrm{Frac}(R_{\mathrm{sec}}) = N.$$

Thus R_{sec}/B is finite free and a hypersurface, hence a complete intersection. Its normalization in N is the already-defined ring T. Put

$$J_{\mathrm{sec}} := f'_{\mathrm{sec}}(\alpha_{\mathrm{sec}}), \qquad \mathfrak{f}_{\mathrm{sec}} := (R_{\mathrm{sec}} :_N T) = \{a \in N : aT \subseteq R_{\mathrm{sec}}\}.$$

Separability gives $J_{\mathrm{sec}} \neq 0$. It is the hypersurface-Jacobian generator of the different [The26g]. The ideal $\mathfrak{f}_{\mathrm{sec}}$ is, by definition, the conductor of the finite normalization $R_{\mathrm{sec}} \subseteq T$. Here the subscript records the order's role in the secant–trace comparison; the element α_{sec} is not asserted to be generated by the secant–frame coefficients, and J_{sec} is neither δ_F nor the Keller constant. No such identification is needed for the reduction below.

Proposition 3.37 (Monogenic Tate–conductor reduction)**.** *For $0 \le j < d$, set*

$$h_j(Z) := \sum_{k=j+1}^{d} \gamma_k Z^{k-1-j}.$$

Then

$$\frac{f_{\mathrm{sec}}(X) - f_{\mathrm{sec}}(Y)}{X - Y} = \sum_{j=0}^{d-1} h_j(X)Y^j,$$

the elements $h_j(\alpha_{\mathrm{sec}})$ form a triangular B-basis of R_{sec}, and, for every $z \in N$,

$$\boxed{J_{\mathrm{sec}} z = \sum_{j=0}^{d-1} h_j(\alpha_{\mathrm{sec}})\, \mathrm{Tr}_{N/K}(z\alpha_{\mathrm{sec}}^j).} \tag{$\mathrm{Tate}_{\mathrm{sec}}$}$$

Moreover,

$$R_{\mathrm{sec}}^{\dagger} := \{z \in N : \mathrm{Tr}_{N/K}(zR_{\mathrm{sec}}) \subseteq B\} = J_{\mathrm{sec}}^{-1} R_{\mathrm{sec}},$$

and therefore

$$\boxed{T^{\dagger} = J_{\mathrm{sec}}^{-1}\mathfrak{f}_{\mathrm{sec}}.} \tag{$\dagger_{\mathrm{sec}}$}$$

Proof. Define the B-linear Tate functional on the power basis by

$$\sigma_{\mathrm{sec}}\left(\sum_{j=0}^{d-1} b_j \alpha_{\mathrm{sec}}^j\right) := b_{d-1}.$$

After tensoring the power basis with K, extend σ_{sec} uniquely to the corresponding K-linear coefficient functional $N \to K$, and retain the same notation. The displayed divided difference is the one-variable Tate–Scheja–Storch coevaluation tensor. Its reconstruction and trace identities are

$$u = \sum_{j=0}^{d-1} h_j(\alpha_{\mathrm{sec}})\sigma_{\mathrm{sec}}(\alpha_{\mathrm{sec}}^j u), \qquad \mathrm{Tr}_{N/K}(u) = \sigma_{\mathrm{sec}}(J_{\mathrm{sec}} u).$$

Applying the first identity to $J_{\mathrm{sec}} z$ and the trace identity to each $\alpha_{\mathrm{sec}}^j z$ gives ($\mathrm{Tate}_{\mathrm{sec}}$). Equivalently, $J_{\mathrm{sec}}^{-1} h_j(\alpha_{\mathrm{sec}})$ is trace-dual to α_{sec}^j. This proves $R_{\mathrm{sec}}^\dagger = J_{\mathrm{sec}}^{-1} R_{\mathrm{sec}}$ [SS75, The26r, The26c].

Finally,

$$T^\dagger = (R_{\mathrm{sec}}^\dagger :_N T) = (J_{\mathrm{sec}}^{-1} R_{\mathrm{sec}} :_N T) = J_{\mathrm{sec}}^{-1}(R_{\mathrm{sec}} :_N T),$$

which is ($\dagger_{\mathrm{sec}}$). Indeed, the first equality follows directly from

$$zT \subseteq R_{\mathrm{sec}}^\dagger \quad \Longleftrightarrow \quad \mathrm{Tr}_{N/K}(zTR_{\mathrm{sec}}) \subseteq B \quad \Longleftrightarrow \quad \mathrm{Tr}_{N/K}(zT) \subseteq B,$$

using $R_{\mathrm{sec}} \subseteq T$ and $1 \in R_{\mathrm{sec}}$. □

Missing morphism: Secant–Frame Trace Landing. The explicit first-jet research target is the ideal containment

$$\boxed{\mathfrak{d}_{C,1}^{\mathrm{sf}} \subseteq \mathfrak{c}_C^{\mathrm{tr}}} \tag{SFTL$_C$}$$

for every prime-order $C \leq G$. Equivalently, every secant–frame denominator must satisfy the uniform trace identity

$$\mathrm{Tr}_{N/K}(srt) \in B \qquad (s \in \mathfrak{d}_{C,1}^{\mathrm{sf}},\ r \in R_C,\ t \in T).$$

Proposition 3.37 gives the equivalent conductor form

$$\boxed{J_{\mathrm{sec}}\, \mathfrak{d}_{C,1}^{\mathrm{sf}} R_C \subseteq \mathfrak{f}_{\mathrm{sec}}} \quad \Longleftrightarrow \quad \mathfrak{d}_{C,1}^{\mathrm{sf}} \subseteq \mathfrak{c}_C^{\mathrm{tr}}. \tag{SCTL$_C$}$$

Because the left-hand side is already a T-submodule, containment in the conductor is equivalently

$$J_{\mathrm{sec}}\, \mathfrak{d}_{C,1}^{\mathrm{sf}} R_C \subseteq R_{\mathrm{sec}}.$$

This also makes the finite part of the reduction explicit. Since T is finite over B, it is finite over R_{sec}. Choose R_{sec}-module generators $t_1, \ldots, t_m$ of T. For $s \in T$ and $r \in R_C$,

$$\begin{aligned} sr \in T^\dagger &\Longleftrightarrow J_{\mathrm{sec}} sr \in \mathfrak{f}_{\mathrm{sec}} \\ &\Longleftrightarrow J_{\mathrm{sec}} srt_i \in R_{\mathrm{sec}} \quad (1 \leq i \leq m) \\ &\Longleftrightarrow \mathrm{Tr}_{N/K}(srt_i \alpha_{\mathrm{sec}}^j) \in B \quad (1 \leq i \leq m,\ 0 \leq j < d). \end{aligned}$$

The final equivalence follows from ($\mathrm{Tate}_{\mathrm{sec}}$) and trace duality. Thus Tate reduction replaces the universal test element $t \in T$ by one fixed finite family of trace coordinates. It does not prove those coordinates integral uniformly for the arbitrary boundary section $r \in R_C$; that uniform-in-r conductor landing is precisely the remaining morphism.

Proposition 3.38 (Scalar landing collapses the boundary overring)**.** *If $0 \neq s \in \mathfrak{c}_C^{\mathrm{tr}}$, then*

$$R_C = T, \qquad \mathcal{P}_C^{\mathrm{fm}} = 0.$$

Equivalently, the same conclusion follows from a nonzero $s \in T$ such that

$$J_{\mathrm{sec}} s R_C \subseteq R_{\mathrm{sec}}.$$

In particular no height-one prime of T contains $\mathfrak{J}_C$.

Proof. The conductor identity converts $sR_C \subseteq T^\dagger$ into $J_{\mathrm{sec}} s R_C \subseteq \mathfrak{f}_{\mathrm{sec}}$, hence into the displayed containment in $R_{\mathrm{sec}} \subseteq T$. Put $a = J_{\mathrm{sec}} s \neq 0$. For every $r \in R_C$ and every $n \geq 0$, the ring property of R_C gives $r^n \in R_C$, and therefore

$$ar^n \in T.$$

Thus every $r \in R_C \subseteq \mathrm{Frac}(T)$ is almost integral over T. A Noetherian domain has the property that almost integral elements of its fraction field are integral; normality then places them in the domain [The26m]. Hence $R_C \subseteq T$. The reverse inclusion is built into the open-complement sequence, so $R_C = T$ and $\mathcal{P}_C^{\mathrm{fm}} = R_C/T = 0$. Statement I now gives the final height-one assertion. □

Assuming (SFTL_C), the nonzero-denominator proposition supplies $0 \neq s_C^{\mathrm{sf}} \in \mathfrak{d}_{C,1}^{\mathrm{sf}} \subseteq \mathfrak{c}_C^{\mathrm{tr}}$. Multiplication then defines the required map explicitly by

$$\lambda_{C,s_C^{\mathrm{sf}}}([r]) = [s_C^{\mathrm{sf}} r].$$

Proposition 3.38 then gives the global chain

$$0 \neq s_C^{\mathrm{sf}}, \quad J_{\mathrm{sec}} s_C^{\mathrm{sf}} R_C \subseteq R_{\mathrm{sec}} \quad \Longrightarrow \quad R_C = T \quad \Longrightarrow \quad \mathcal{P}_C^{\mathrm{fm}} = 0.$$

The local bounded-pole calculation below remains the geometric explanation for why this is impossible at a hidden divisor, but is not needed for this global implication.

Merely asking for a nonzero element of $\mathfrak{d}_{C,1}^{\mathrm{sf}} \cap \mathfrak{c}_C^{\mathrm{tr}}$ would not define a new mechanism: since T is a domain and $\mathfrak{d}_{C,1}^{\mathrm{sf}} \neq 0$, that intersection is nonzero exactly when $\mathfrak{c}_C^{\mathrm{tr}} \neq 0$, which is boundary separation itself. In fact the dichotomy below shows that the full containment (SFTL_C) is logically equivalent to the same separation assertion. Its value as a research target is therefore not a weaker logical premise, but the prescribed coefficientwise trace identity that one can try to prove directly from the displayed secant–frame data. No identity proved here yet shows that first jets suffice; jet order (1) is the first explicitly typed candidate.

More generally, write (SFTL_C) = ($\mathrm{SFTL}_{C,1}$), and for every finite $\ell \geq 0$ state

$$(\mathrm{SFTL}_{C,\ell})\colon \mathfrak{d}_{C,\ell}^{\mathrm{sf}} \subseteq \mathfrak{c}_C^{\mathrm{tr}}.$$

Any one finite-order statement suffices by the same argument, whereas ($\mathrm{SFTL}_{C,1}$) is the designated first-jet proposal. A minimal scalar certificate would instead specify, by an explicit secant–frame formula, one $0 \neq s_{C,\ell}^{\mathrm{sf}} \in \mathfrak{d}_{C,\ell}^{\mathrm{sf}}$ and prove its trace landing. Merely asserting that some nonzero element lies in the intersection is not a new mechanism; the formula producing that element is essential. The ideal form used above is choice-free and asks for landing coefficientwise.

The nonvanishing proposition is only finite denominator clearing; it does not prove landing for the generally nonfinite overring R_C. Indeed, if a height-one prime $\mathfrak{p}$ contains $\mathfrak{J}_C$, Statement I and the localized open-complement sequence, together with Statement II, give

$$(R_C)_{\mathfrak{p}} = \mathrm{Frac}(T_{\mathfrak{p}}), \qquad (T^{\dagger})_{\mathfrak{p}} = \pi^{-(e_{\mathfrak{p}}-1)}T_{\mathfrak{p}}.$$

For every $0 \neq s \in T$, multiplication by $s/1$ on $\mathrm{Frac}(T_{\mathfrak{p}})/T_{\mathfrak{p}}$ is surjective and therefore cannot land in the proper bounded stage $\pi^{-(e_{\mathfrak{p}}-1)}T_{\mathfrak{p}}/T_{\mathfrak{p}}$. Consequently $(\mathfrak{c}_C^{\mathrm{tr}})_{\mathfrak{p}} = 0$. Since $T \to T_{\mathfrak{p}}$ is injective, this forces $\mathfrak{c}_C^{\mathrm{tr}} = 0$, and (SFTL$_C$) excludes $\mathfrak{p}$. Notice that no unit or avoidance condition $s_C^{\mathrm{sf}} \notin \mathfrak{p}$ is needed: global nonzeroness already survives localization in the domain T.

Conversely, boundary separation gives $R_C = T$, hence $\mathfrak{c}_C^{\mathrm{tr}} = T$. Thus nonzero trace landing is an exact Keller-specific formulation of the open divisorial assertion, rather than a consequence of finite flatness. The secant and frame data now provide a concrete finite candidate ideal; the missing theorem is the uniform passage from those finitely many coefficients to every $r \in R_C$.

Thus the purity argument remains the unique endgame. Statements I and II identify its unbounded source and finite target, while (SFTL$_C$) is a concrete first-jet construction problem whose solution would establish its height-one premise.

Lean correspondence. The stable API formalizes the canonical ideals `fixedLocusIdeal`, `movingBoundaryIdeal`, and `fixedMovingBoundaryIdeal`, together with `PlanarBoundarySeparation`, the pulled-back conjugate boundaries, their valuation-theoretic containment at ramified divisors, and the ensuing purity endgame. It takes planar Ax–Grothendieck, branch purity, finite- étale rigidity, and a supplied `PlanarKellerCollisionModelF` as explicit interfaces. The Lean separation predicate currently quantifies over every nontrivial subgroup; formalizing its prime-order Cauchy reduction is a remaining routine API task.
The general object `BoundaryPrincipalParts` and its canonical specialization `FixedMovingBoundaryPrincipalParts` are defined. The local-cohomology localization, DVR computation, and S_2-vanishing used in Statement I are not yet formalized. The research layer defines the generic `TraceIntegralSubmodule` and `boundedStageComparisonIdeal`; it proves the former equal to Mathlib's `Submodule.traceDual` on the unit order. The opt-in modules `Planar.Research.TateReconstruction` and `Planar.Research.MonogenicTraceDual` now formalize the finite Tate sum, the monogenic trace-dual formula, and the overorder conductor–codifferent identity. Trace-dual/Hom finiteness and the local inverse-different assertion of Statement II remain unfinished in the planar specialization. Secant evaluation on a generic collision cocone is formalized in `CollisionIdeals/Planar/ConjugateSecantEvaluation.lean`; the actual g-sheet specialization and coefficient family remain missing. The choice-free telescoping coefficients and polynomial Keller frame are in `Planar.ExplicitSecant` and `Planar.KellerFrame`. The research theorem `finiteCoefficientDenominatorIdeal_ne_bot` formalizes finite denominator clearing, but the actual conjugate evaluated family is missing. The opt-in module `Planar.Research.MonogenicOrder` directly reuses Mathlib's `Algebra.adjoin`, `Algebra.adjoin.powerBasis'`, `minpoly.equivAdjoin`, and `conductor` for the monogenic comparison order. The module `Planar.Research.MonogenicLanding` defines the conductor bounded stage and its transporter and proves an abstract landing-to-endgame implication from a supplied pole witness. The open-section algebra R_C, its common embedding with the bounded stage in N/T, the concrete conjugate evaluated secant–frame family, the specialization of the generic finite-overring theorem to R_C, and the ideal containment (SFTL$_C$) remain to be formalized. All these research objects remain behind `CollisionIdeals/Planar/Research.lean`, outside the stable import spine.

Remark (Exact remaining content of classical $JC(2)$). The traditional complex conjecture $JC(2)$ has no boundary, coverage, or module-finiteness condition in its definition: it asserts simply that every planar Keller map is a polynomial automorphism. Corollary 3.33 proves that its universal content is equivalent to the boundary formulations above. Theorem 3.15 settles every normal marked extension, and Corollary 3.16 settles generic degree at most two. Consequently, to prove $JC(2)$ it remains to show that every planar Keller map with L/K nonnormal and $d_F \geq 3$ satisfies

$$\mathrm{codim}_Z\, V(\mathfrak{a}_C + \mathfrak{j}_C^{\mathrm{mov}}) \geq 2 \quad \text{for every prime-order } C \leq \mathrm{Gal}(N/K).$$

Equivalently, one must exclude a common height-one fixed–moving boundary component in the normal closure of every remaining nonnormal extension. This is an exact reformulation of the unresolved part of $JC(2)$, not an additional hypothesis silently built into the classical conjecture.

4 The cubic S_3 obstruction in dimension three

We now specialize the dimension-independent constructions of Section 2 to dimension three and isolate the separable nonnormal cubic case.

Let $S_3 = \mathrm{Perm}(\{1, 2, 3\})$ be the symmetric group on three letters.

Proposition 4.1 (Dimension-three counterexample criterion)**.** *For a polynomial map*

$$F: \mathbb{A}^3_{\mathbb{C}} \longrightarrow \mathbb{A}^3_{\mathbb{C}},$$

consider the following statements:

(1) $\det JF \in \mathbb{C}^\times$*;*

(2) *F is not a polynomial automorphism;*

(3) $R^\circ_F \neq \varnothing$*.*

Statements (2) *and* (3) *are equivalent. Consequently, F is a dimension-three Jacobian-conjecture counterexample exactly when* (1) *and* (3) *hold.*

Proof. By Corollary 2.4 and Theorem 2.15,

$$R^\circ_F = \varnothing \quad \Longleftrightarrow \quad I_R = I_\Delta \quad \Longleftrightarrow \quad F \in \mathrm{Aut}_{\mathrm{poly}}(\mathbb{A}^3_{\mathbb{C}}).$$

Negating this equivalence proves (2)$\Longleftrightarrow$(3). Combining this with (1) gives the final assertion. $\square$

4.1 The cubic extension and its normal closure

Let

$$F = (F_1, F_2, F_3): \mathbb{A}^3_{\mathbb{C}} \longrightarrow \mathbb{A}^3_{\mathbb{C}}$$

be dominant and generically finite. Let

$$A = \mathbb{C}[x_1, x_2, x_3], \qquad B = \mathbb{C}[F_1, F_2, F_3] \subseteq A, \qquad K = \mathrm{Frac}(B), \qquad L = \mathrm{Frac}(A),$$

and let

$$\theta: K \otimes_B A \longrightarrow L, \qquad \lambda \otimes a \longmapsto \lambda a,$$

be the isomorphism of Theorem 2.6. Suppose that L/K is separable and nonnormal, and choose a power basis with generator α and degree three:

$$L = K(\alpha), \qquad [L : K] = 3.$$

If $m_\alpha \in K[T]$ is the minimal polynomial of α, let $h_\alpha \in L[T]$ be determined by

$$m_\alpha(T) = (T - \alpha)h_\alpha(T) \qquad \text{in } L[T].$$

Let N/K be the normal closure of L/K, containing L through the marked inclusion. Let

$$G = \mathrm{Gal}(N/K), \qquad H = \mathrm{Gal}(N/L).$$

Recall from Section 2.5 that

$$G/H \xrightarrow{\sim} \mathrm{Hom}_K(L, N), \qquad gH \longmapsto g|_L, \qquad \mathrm{core}_G(H) = 1.$$

The final equality follows because N is the normal closure of L/K; equivalently, the action of G on the conjugate embeddings of L is faithful.

For $Y = \operatorname{Spec} B$, the finite-normalization construction of Section 2.5 gives

$$Z = \mathrm{Norm}_N(Y) \longrightarrow \overline{X} = \mathrm{Norm}_L(Y) \longrightarrow Y.$$

If $E \in Z^{(1)}$, with decomposition group D_E and inertia group I_E, then its conjugate centers are indexed by $\mathcal{Q}_E = D_E\backslash G/H$. For $\omega = D_E gH$, Proposition 2.11 states

$$e\big(\mathrm{cent}(E,\omega)/\nu(E)\big) = i_E(\omega) = [I_E : I_E \cap gHg^{-1}].$$

The following proposition is the standard residual-factor description of a separable nonnormal cubic extension. We include its short proof to fix the marked identification with N.

Proposition 4.2 (Cubic residual-factor identification)**.** *The polynomial h_α is irreducible over L, and there is an L-algebra isomorphism*

$$\varphi\colon L[T]/(h_\alpha) \xrightarrow{\sim}_L N.$$

Consequently,

$$\Psi\colon L \otimes_K L \xrightarrow{\sim}_L L \times N, \qquad \mathrm{pr}_1 \circ \Psi = \mu_L.$$

Proof. Write

$$m_\alpha(T) = T^3 + a_2T^2 + a_1T + a_0.$$

If the quadratic h_α were reducible over L, it would have a root $\beta \in L$. Separability gives $\beta \neq \alpha$, and the third root would be

$$\gamma = -a_2 - \alpha - \beta \in L.$$

Thus m_α would split over L. Since $L = K(\alpha)$, the extension L/K would then be the splitting field of the separable polynomial m_α, hence normal. This contradicts the assumed nonnormality, so h_α is irreducible.

Let β denote the image of T in $L[T]/(h_\alpha)$. This field contains the three roots

$$\alpha, \qquad \beta, \qquad -a_2 - \alpha - \beta$$

of m_α, and is generated over L by β. It is therefore the splitting field of m_α, which is the normal closure N of L/K. Choose the resulting L-algebra isomorphism

$$\varphi\colon L[T]/(h_\alpha) \xrightarrow{\sim}_L N.$$

Let

$$\Phi_\alpha\colon L \otimes_K L \xrightarrow{\sim}_L L \times L[T]/(h_\alpha)$$

be the marked-root decomposition of Theorem 2.7. It satisfies

$$\mathrm{pr}_1 \circ \Phi_\alpha = \mu_L.$$

Composing its second factor with φ, let

$$\Psi = (\mathrm{id}_L \times \varphi) \circ \Phi_\alpha.$$

Since $\mathrm{id}_L \times \varphi$ leaves the first factor unchanged, $\mathrm{pr}_1 \circ \Psi = \mu_L$. □

Theorem 4.3 (Galois group of a nonnormal cubic extension)**.**

$$\mathrm{Gal}(N/K) \cong S_3.$$

Proof. Proposition 4.2 identifies N with the degree-two extension $L[T]/(h_\alpha)$ of L. Hence

$$|H| = [N : L] = 2.$$

The normal-closure identities recalled above give

$$\mathrm{core}_G(H) = 1, \qquad [G : H] = [L : K] = 3.$$

Hence the action of G on the three cosets of H gives

$$\rho\colon G \longrightarrow \mathrm{Perm}(G/H) \cong S_3, \qquad \ker\rho = \mathrm{core}_G(H) = 1.$$

Core-freeness therefore makes ρ injective, so

$$|G| \le |S_3| = 6.$$

The index formula gives

$$|G| = [G : H]|H| = 3 \cdot 2 = 6.$$

Thus $|G| = 6$, and the injective homomorphism ρ is an isomorphism $G \cong S_3$. □

4.2 The cubic off-diagonal factor

Retain the cubic data of Section 4.1, and suppose in addition that F is Keller. Thus

$$B = \mathbb{C}[F_1, F_2, F_3] \subseteq A = \mathbb{C}[x_1, x_2, x_3], \qquad K = \mathrm{Frac}(B) \subseteq L = \mathrm{Frac}(A) = K(\alpha) \subseteq N,$$

where L/K is separable, nonnormal, and of degree three, and N/K is its normal closure. Let

$$\theta\colon K \otimes_B A \xrightarrow{\sim} L, \qquad \lambda \otimes a \longmapsto \lambda a.$$

Corollary 4.4 (Generic cubic S_3 collision)**.** *There is a K-algebra isomorphism*

$$\Psi_F\colon K \otimes_B C_F \xrightarrow{\sim}_K L \times N, \qquad \mathrm{pr}_1 \circ \Psi_F = \theta \circ (1 \otimes \bar{\mu}_F).$$

It satisfies

$$\Psi_F(\ker(\theta \circ (1 \otimes \bar{\mu}_F))) = 0 \times N, \qquad \mathrm{Gal}(N/K) \cong S_3.$$

Consequently,

$$\mathrm{Obs}(F) \neq 0, \qquad I_R \subsetneq I_\Delta, \qquad R_F^\circ \neq \varnothing, \qquad F \notin \mathrm{Aut}_{\mathrm{poly}}(\mathbb{A}^3_{\mathbb{C}}).$$

Proof. Theorem 2.6 gives a K-algebra isomorphism

$$\Phi_F \colon K \otimes_B C_F \xrightarrow{\sim}_K L \otimes_K L$$

such that

$$\mu_L \circ \Phi_F = \theta \circ (1 \otimes \bar{\mu}_F).$$

Proposition 4.2 gives

$$\Psi \colon L \otimes_K L \xrightarrow{\sim}_L L \times N, \qquad \mathrm{pr}_1 \circ \Psi = \mu_L.$$

After restricting Ψ from L to K, let

$$\Psi_F = \Psi \circ \Phi_F.$$

It follows that

$$\mathrm{pr}_1 \circ \Psi_F = \theta \circ (1 \otimes \bar{\mu}_F), \qquad \Psi_F\big(\ker(\theta \circ (1 \otimes \bar{\mu}_F))\big) = 0 \times N \neq 0.$$

Suppose that the diagonal-evaluation homomorphism $\bar{\mu}_F \colon C_F \to A$ were injective. Since K is a localization of B, it is flat over B, so $1 \otimes \bar{\mu}_F$ would also be injective. Theorem 2.6 shows that θ is an isomorphism. Therefore $\theta \circ (1 \otimes \bar{\mu}_F)$ would be injective, contradicting the nonzero kernel $0 \times N$ above. Hence $\ker(\bar{\mu}_F) \neq 0$, and Proposition 2.2 gives $\mathrm{Obs}(F) \neq 0$. Proposition 2.1 shows that $I_R \subseteq I_\Delta$ strict, and Corollary 2.4 gives $R_F^\circ \neq \varnothing$.

Finally, Theorem 4.3 identifies $\mathrm{Gal}(N/K)$ with S_3. If F were a polynomial automorphism, Theorem 2.15 would give $\mathrm{Obs}(F) = 0$, contrary to $\mathrm{Obs}(F) \neq 0$. Since F is Keller, it is a dimension-three Jacobian-conjecture counterexample. $\square$

For the same cubic map and field tower, let

$$X = \operatorname{Spec} A, \qquad Y = \operatorname{Spec} B, \qquad G = \mathrm{Gal}(N/K), \qquad H = \mathrm{Gal}(N/L).$$

Suppose there is a diagram over Y

$$\mathcal{D} = \Big(Z = \mathrm{Norm}_N(Y) \xrightarrow{\pi} \overline{X} = \mathrm{Norm}_L(Y) \xrightarrow{\bar{\nu}} Y,\ j \colon X \hookrightarrow \overline{X}\Big)$$

in which π and $\bar{\nu}$ are finite, j is an open immersion, $\bar{\nu} \circ j = f_F$, and $f_F \colon X \to Y$ is étale. Let

$$\partial X = \overline{X} \setminus j(X), \qquad G \cong S_3$$

by Theorem 4.3.

Corollary 4.5 (Hidden inertia in the S_3-normalization)**.**

$$\mathcal{H}(\mathcal{D}) = \mathrm{Ram}^{(1)}(Z/Y).$$

Proof. Apply Proposition 2.14 to the displayed normalization diagram. $\square$

Remark (Ordered roots)*.* Let

$$m_\alpha(T) = T^3 + a_2 T^2 + a_1 T + a_0$$

have roots $\alpha_1, \alpha_2, \alpha_3$ in N. For distinct i, j, k,

$$\alpha_k = -a_2 - \alpha_i - \alpha_j, \qquad K(\alpha_i, \alpha_j) = K(\alpha_1, \alpha_2, \alpha_3) = N.$$

Thus the off-diagonal factor, which marks two distinct roots, is already the ordered-root normal closure.

Lean correspondence. `CollisionIdeals/ComplexThree/CubicGaloisGroup.lean`, `CollisionIdeals/ComplexThree/S3Collision.lean`, `CollisionIdeals/ComplexThree/OffDiagonal.lean`, `CollisionIdeals/ComplexThree/CubicS3.lean`, and `CollisionIdeals/ComplexThree/JacobianConjecture.lean`. The Lean cubic witness takes the residual-field isomorphism and $H \neq 1$ as explicit inputs; Proposition 4.2 derives them here from nonnormality. Hidden inertia is formalized pointwise at a supplied ramified divisor.

Remark (Exact relation with classical complex $JC(3)$). The traditional complex three-dimensional Jacobian conjecture asserts that every polynomial map

$$F\colon \mathbb{A}^3_{\mathbb{C}} \longrightarrow \mathbb{A}^3_{\mathbb{C}}, \qquad \det JF \in \mathbb{C}^{\times},$$

is a polynomial automorphism; equivalently, every such map satisfies $\mathrm{Obs}(F) = 0$. Corollary 4.4 proves the following implication under a prescribed function-field hypothesis:

$$\begin{gathered} \det JF \in \mathbb{C}^{\times}, \qquad [L:K]=3, \qquad L/K \text{ nonnormal} \\ \Longrightarrow \quad \mathrm{Obs}(F) \neq 0 \quad \Longrightarrow \quad F \text{ is a counterexample to } JC(3). \end{gathered}$$

Consequently, the existence of one such Keller map would disprove $JC(3)$, while $JC(3)$ implies that no such map exists. This section does not construct such a map, and it does not assert that every possible counterexample to $JC(3)$ has generic degree three. It identifies the generic collision geometry of this cubic nonnormal subclass.

5 Conclusion

The obstruction ideal $\mathrm{Obs}(F)$ vanishes exactly when the scheme-theoretic collision relation $X \times_Y X$ equals the diagonal. In dimension two there is one endgame with several exact formulations. PlanarBoundaryCoherence expresses it through disappearance of the whole deleted boundary. PlanarRamificationRigidity, moving-sheet coverage, and boundary separation express its height-one premise in differential, valuation-theoretic, and finite-group language. Only after that premise is established do purity and finite-étale rigidity make the normal closure trivial and force $\mathrm{Obs}(F) = 0$. The image of the deleted boundary is exactly the nonproperness set, and the boundary divisor-class argument proves the normal/Galois case unconditionally. Universal validity of the equivalent planar criteria is exactly $JC(2)$, not a result asserted here. In the cubic dimension-three class, by contrast, the nonzero generic factor is the ordered-root S_3-normal closure. Both statements arise from the same collision, off-diagonal, and conjugate-sheet constructions; their point is to locate the obstruction, respectively, at the normalization boundary and on the generic off-diagonal sheet.

Under PlanarBoundaryCoherence or PlanarRamificationRigidity in dimension two, under the conditions of Section 4 in dimension three, and with $\mathrm{Ram}^{(1)}(Z/Y) \neq \varnothing$ in the cubic normalization, the contrast is

planar rigidity criteria	nonnormal cubic class in dimension three
$\mathrm{PBC}(F)$ or $\mathrm{PRR}(F)$	hidden inertia in the S_3-cover realized
$\mathrm{Obs}(F) = 0$	$\mathrm{Obs}(F) \neq 0$
$I_R = I_\Delta$	$I_R \subsetneq I_\Delta$
F is a polynomial automorphism	F is not a polynomial automorphism.

Motivating example. The explicit three-dimensional polynomial counterexample announced by Levent Alpöge [Alp26] and independently formalized in Isabelle/HOL by Ramos, Hulak, and de Queiroz [RHdQ26] motivated the cubic S_3 comparison. The formal verification establishes a constant nonzero Jacobian determinant and distinct rational points with the same image. That particular map is not used in the arguments above: the dimension-three result is the structural implication from the stated cubic and marked-normal-closure conditions.

The formalization uses Lean 4 and Mathlib [dMU21, The20].

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